Earth-State Embeddings:

self-supervised representations of the observed ocean, probed against the Atlantic overturning circulation

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Abstract

We train a small self-supervised model to compress everything the observing system knows about each point of the ocean, each month, into a single vector — an *embedding* — and then ask whether those vectors, which never see any transport measurement, carry the Atlantic Meridional Overturning Circulation (AMOC). They do: a linear read-out from the embeddings along 26.5°N attains a year-blocked cross-validated correlation of $r = 0.60$ with the monthly, deseasonalized RAPID transport series — to our knowledge the first cross-validated monthly-scale figure against real RAPID data from inputs that never include RAPID. A temporal model over the frozen embeddings predicts the ocean's anomaly field beyond a year ahead (anomaly correlation 0.62 at one month, crossing the conventional 0.5 useful-skill line near four months), while the transport read-out loses skill after about one season, consistent with its wind-driven monthly variance. This revision adds a *probe ladder* — ridge, small MLP, and a supervised attention head over the *unpooled* section — and it rewrites the story of the week. Pointwise nonlinearity recovers nothing anywhere: the MLP trails the ridge on every target, for every codec. But the attention head, which sees the ∼67 section pixels individually instead of their mean, raises every codec — and the codec whose channel tokens carry 3 × 3 spatial patches reaches $\mathbf{r = 0.672}$ across two independently trained seeds (0.690 and 0.654), RMSE ≈2.1 Sv against a 2.79 Sv target, the best transport figure of the programme. The mechanism is thermal wind: geostrophic transport is the east-minus-west density difference across the section, and mean-pooling computes exactly the statistic in which that difference cancels. An end-to-end control then decomposes the headline: the same head over *raw* section data reaches 0.659 at matched receptive field, so pretraining's own margin on this task is +0.013 — indistinguishable from zero, and a directional +0.031 reported from the first seed alone is withdrawn here — the codec's demonstrated value lies in field prediction and transfer, not in the monthly RAPID probe. An apparent finding that expanding the input from 14 to 24–25 channels had *cost* monthly RAPID skill is retracted as a representation claim — the pooled probes were billing relocated information as lost — while the genuine gain those channels brought at the deep MOVE array at 16°N ($r = 0.206$ [0.044, 0.346], our first multi-target confidence interval excluding zero) stands.

This revision (v6) reports the programme's move to the *rolled forecast at scale* (§8), and its central finding is about capacity. The temporal model now reads a *stencil* of distant neighbour embeddings alongside the pixel's own history; skill is scored both one step ahead (forecast ratio against persistence) and rolled twelve months over the AMOC corridor (corridor AUC), with a frozen gate head that must reproduce its pinned score before any evaluation counts. At the original stage-2 size (576 × 8, 34M parameters) the two scores *anti-correlated* across input geometries — the best one-step arms rolled worst — and a systematic audit cleared the instrument and localized the effect in the model-under-iteration. The resolution is capacity: at 768 × 12 (88M) the same wide-input geometry that lost at 34M gains +0.050 corridor AUC paired by seed, and at 1024 × 16 (205M) the corridor AUC reaches **0.664**/0.663/0.664 across three seeds — against 0.589 for the no-neighbour baseline and 0.604 for the best small-model geometry — with no sign of saturation. On the one-step forecast, parameters beat every other axis measured: input width (Fibonacci sunflower layouts, 34→89 points, ≈ −0.002 per rung, then *saturating* at 89 — 144 points buy nothing) and step budget (60k → 200k, decelerating toward an asymptote) both pay, and both pay less. **On the rolled forecast, scale is not the largest effect, and neither axis behaves as the forecast ratio said it would**: the step budget that bought the most one-step skill buys *exactly nothing* rolled, while the width that had visibly saturated at 89 points keeps paying to 144 (0.6637 → 0.6725 → 0.6781 at 205M, same sign at every seed) and then stops, a paired 233-point stencil returning 0.6739. Adding Gaussian noise to the model's own inputs at its measured one-step error scale is worth +0.057 corridor AUC at fixed capacity — enough that a noised 88M model beats the clean 205M one — and a near-field 8-point stencil is worth +0.035 over wide-and-far inputs at

equal parameters, reversing the forecast-axis ranking. A geometric measurement plausibly explains both: at 4444 km reach a head's dependency cone covers the entire model domain by horizon three and about half its input slots fall outside it, so every rolled number here rests on holding the world beyond the window at climatology. A second structural caveat is newly located and quantified: 45°W–25°W is withheld from training in both stages and scored anyway, it is a quarter of the evaluation area, and roughly half of the stencil's measured rolled advantage is earned inside it — so *spatial coupling helps forecasting* and *spatial coupling helps a model extrapolate into a hole in its own coordinate input* are not separated by any number in this report (§8.1). The small-model geometry ranking that occupied a week of experiments is preserved in Appendix A as a capacity-starvation artifact.

1 Introduction

The AMOC moves on the order of 17 Sv of water northward through the Atlantic's upper kilometre ($1\,\mathrm{Sv} = 10^6\,\mathrm{m}^3\,\mathrm{s}^{-1}$), carries roughly a petawatt of heat, and is directly measured at only a handful of latitudes, for only about two decades. It is the canonical example of a climate-critical quantity that is *composite*: part wind-driven (Ekman transport), part density-driven (geostrophic shear), part boundary currents. No single satellite or float sees it; it must be inferred from the state of the ocean around it.

This project asks a representation-learning question of that inference problem: *if we compress the whole observable ocean state into generic, task-blind embeddings, do climate-critical integrals like the AMOC come along for free?* The design is deliberately strict: the embedding model never sees any transport data; every read-out is a linear (ridge) regression, so any skill must already be present, linearly accessible, in the representation; and every claim is cross-validated in blocks that respect the autocorrelation of the climate system. The project also serves as a live test of a foundation-model hypothesis (§6): that a single representation trained on *all* the data — every ocean, every channel — serves specific regional tasks at least as well as one trained only on the task's own region.

2 Terminology

Terms are used throughout in the following precise senses.

Transport the physical quantity all "truth" targets measure: the volume of water a current system moves, in Sverdrups ($1\,\mathrm{Sv} = 10^6\,\mathrm{m}^3\,\mathrm{s}^{-1}$). The AMOC's strength is a transport (∼17 Sv northward across 26.5°N). A *transport series* is one array's monthly record of it; none is ever a model input.

Fold one round of the year-blocked cross-validation: a calendar year serves as the test set while the probe is re-fitted from scratch on all other years. "Per fold" means exactly that re-fitting — no test month ever influences the weights that predict it.

Transport claims / field-prediction claims the project's two scoreboards, kept apart because their baselines differ. Transport claims: predicting the mooring series (k-fold r, RMSE in Sv, event capture) against the wind-only ridge. Field-prediction claims: predicting the ocean's own input fields ahead in time (chan%, rollout MSSS/ACC) against persistence and climatology.

Channel one physical input field: e.g. surface current speed, mixed-layer depth, temperature at 700 dbar, zonal wind stress. The data tensor has 12–15 channels in the first generation, 24–25 in the second, and 39 in the third (Table 1, §3.2).

Tensor generation one build of the input tensor whose channel set, grid, or time axis differs from the others enough that results are not comparable across it (§3.2). Three exist, and we name them by what defines them: the **coarse tensor** (1°, 4 Argo pressure levels; "family 1" in earlier notes), the **deep tensor** (1°, 8 Argo levels; "family 2"), and the **quarter-degree tensor** (0.25° North Atlantic, 16 Argo levels, sea-surface height added, time axis extended to 1982; "family 3"). Run names keep the short prefixes (`f3_` etc.); prose uses the descriptive names.

Pixel one ocean grid cell — 1° × 1° in the first two tensor generations, 0.25° × 0.25° in the third. A *pixel-month* is one pixel at one month — the atomic sample.

Channel space the space of the physical fields themselves. An error "in channel space" is an error in predicted physical values (normalized units), directly comparable with a persistence forecast of the same fields.

Embedding / embedding space the 64-dimensional vector z the encoder produces for a pixel-month, and the space of such vectors. An error "in embedding space" compares predicted to actual z; it is an internal diagnostic, since z has no physical units.

Codec the stage-1 encoder–decoder (§4); it encodes a pixel-month's channels to z and can decode z back to channels. "Codec" emphasizes that it is a learned compression.

Anomaly space all dynamic channels are expressed as departures from that pixel's own monthly climatology (computed over training years only), then normalized. This removes everything predictable from location and calendar alone.

Climatology the long-term mean for a given pixel and calendar month. As a forecast, "climatology" predicts zero anomaly — the no-skill floor every real forecast must beat.

Persistence the forecast that the anomaly stays exactly as it is now. Hard to beat at short range; decays into a poor forecast at long range. *Damped persistence* multiplies the anomaly by its own lag-h autocorrelation — the fair statistical baseline.

Patch codec a codec variant whose channel token carries the 3×3 neighbourhood of the pixel (nine values and nine observed-flags) rather than a single scalar, giving the encoder gradient vision and turning the representation from an expander into a genuine compressor (§4).

Stage 2 / temporal model a causal transformer over a pixel's sequence of frozen embeddings that predicts next month's embedding. "Frozen" means the codec's weights are not updated by stage-2 training, so improvements are attributable.

Probe a read-out fitted from frozen embeddings to a target series; the codec and stage 2 are never updated by it. Historically a ridge regression, deliberately weak; now the bottom rung of the probe ladder below. A probe measures what a representation *contains*.

Probe ladder three read-outs of the same frozen embeddings, increasing only in what they may see: *ridge* (linear, on the section-mean embedding), *MLP* (adds pointwise nonlinearity, same pooled input), *attention head* (adds spatial structure: one learned query attends over the unpooled per-pixel embeddings of the section). Each rung localizes where a capability comes from.

Section the row of grid cells nearest a mooring array's latitude, clipped to the array's longitude span (e.g. RAPID: 26.5°N, 80°W–13°W). Probes read the mean embedding along the target's section.

Holdout years entire calendar years (2009, 2017, 2023) excluded from all training; with the held-out longitude block below they form the blocked validation splits. Random splits are never used — adjacent months are too correlated to count as independent.

Holdout longitudes the meridional block 45°W–25°W (`train.py --holdout-lon`, half-open in longitude), excluded from training in *both* stages: the codec never fits a pixel there and the forecast head never takes a training window centred there. It is 25.0% of the evaluation window, 23.9% of the corridor and 30.2% of the RAPID section — and the rolled evaluation scores it along with everything else, so every rolled number in this report is a blend of trained- and untrained-longitude skill. §8.1 and Fig. 7 measure what that costs.

k-fold (year-blocked) every calendar year takes one turn as the test set while the probe is fit on the others; the assembled out-of-fold predictions are scored once. Confidence intervals come from a bootstrap that resamples whole years (a *block bootstrap*).

chan% held-out next-month prediction error reduction versus persistence, in channel space, in anomaly space: $100\,(1-\mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{pers}})$. Comparable only within one channel set, since the persistence baseline moves when channels change.

MSSS mean squared skill score, $1 - \mathrm{MSE}/\mathrm{MSE}_{\mathrm{ref}}$ [6]; our rollout reports it against climatology, persistence, and damped persistence.

ACC anomaly correlation coefficient: the Pearson correlation between predicted and observed anomalies at a given forecast horizon. ACC $= 0.5$ is the conventional "useful skill" line.

RAPID / FC / MOVE / OSNAP / SAMBA the five transport truth series: the RAPID array at 26.5°N (2004–, monthly), the Florida Current cable at 27°N (1982–, daily), MOVE at 16°N (2000–2022, the *deep* western-boundary flow), OSNAP in the subpolar gyre (2014–2022), SAMBA at 34.5°S (2009–2017). None is ever an input.

channel(s)	source	record	role
surface current speed, MLD	GLORYS reanalysis	1993–	dynamics
sea-surface height ($\frac{1}{4}$° only)	GLORYS reanalysis	1993–	geostrophy
T at 4 / 8 / 16 pressure levels	Roemmich–Gilson Argo	2004–	density structure
S at 4 / 8 / 16 pressure levels	Roemmich–Gilson Argo	2004–	density structure
wind stress τ_x, τ_y	NCEP R1	1948–	Ekman forcing
within-month τ std. dev.	NCEP R1 daily	1948–	storminess
monthly SST (optional, 1° only)	OISST v2.1	1981–	thermal memory
SST, precip climatologies (1° only)	OISST, GPCP	fixed	(measured inert; §9)

Table 1: Channels of the data tensor across its three generations (§3.2). Coarse and deep tensors: monthly, 1°, 1993–present; NA pilot window (100°W–20°E, 0–70°N; 5,787 ocean pixels) or global (44,964 pixels). Quarter-degree tensor: monthly, 0.25°, NA window, 84,405 ocean pixels, axis extended to 1982-01 (channels absent before their records enter as missing tokens), C=39. Missing data (Argo before 2004, polar gaps, pre-1993 months) is represented explicitly, never imputed.

Stencil the set of *neighbour pixels* whose embedding sequences the temporal model reads alongside the centre pixel's own (§8). A stencil of S slots turns the model's input from one 24-month embedding sequence into S+1 of them. Geometries used: rings at a fixed radius, spirals (every point on its own bearing), and *sunflower* layouts — golden-angle spirals with a far-heavy radial ramp, at Fibonacci counts 34/55/89/144, reaching 111–4444 km. A neighbour that falls on land or off-window is a *dead slot*, encoded as zero.

Forecast ratio the one-step scoreboard of the stencil programme: $\mathrm{MSE}_{\mathrm{model}}/\mathrm{MSE}_{\mathrm{pers}}$ for z_{t+1} prediction on held-out windows, in embedding space. Lower is better; persistence sits at 1.0 by construction and the measured persistence MSE is ≈3.1 z-units^2. Reproduces to ±0.002 across seeds, so paired differences of 0.002 are readable.

Corridor the scored subset of the rolled evaluation: the fastest quarter of the window's pixels by train-month mean current speed, dilated two cells, plus the RAPID section — 29,627 of 84,405 window pixels. It is where the AMOC's limbs live, and it is rendered as a globe layer in the companion visualization.

Corridor AUC the rolled scoreboard: initialize with true embeddings, roll twelve months feeding predictions back, decode, and score MSSS against climatology over the corridor's pixel-months at each horizon,

$$\mathrm{AUC} \;=\; \frac{1}{12}\sum_{h=1}^{12}\Big(1-\tfrac{\mathrm{MSE}_h}{\mathrm{MSE}_{\mathrm{clim},h}}\Big).$$

"AUC" is *area under the skill-versus-horizon curve* — a plain mean over twelve monthly horizons — **not** a ROC statistic. Higher is better; climatology sits at 0 by construction, and the ranking is unchanged under persistence and damped-persistence references and under horizon re-weighting (§8.2). Table 5 is the reference table.

Gate head a frozen reference head (`e017`, no neighbours) re-evaluated at the start of every corridor evaluation; it must reproduce its pinned AUC of 0.643 to the third decimal or the evaluation is void. Every number in §8 passed this gate.

Capacity tiers the three stage-2 sizes of §8, with parameters measured from the checkpoints: *base* 576 × 8 (34.1M at 56 slots), *big* 768 × 12 (88.0M), *xl* 1024 × 16 (205.4M).

3 Data

The tensor (Table 1) is built entirely from credential-free public archives. Two properties matter most. First, *missingness is information*: a pixel-month's unobserved channels enter the encoder as learned "missing" tokens rather than imputed values, so the representation knows what the observing system knew. Second, everything is in anomaly space (§2) — the founding negative result of the project was that a model trained on raw fields wins reconstruction by memorizing the seasonal cycle while learning nothing about change.

The five truth series (RAPID, FC, MOVE, OSNAP, SAMBA) ride along as never-input targets: 240 RAPID months plus 792 additional truth months across four other latitudes.

3.1 The sources, one by one

Every input is a credential-free public archive; this subsection says what each one is, what channels it contributes, and why it is in the tensor. (The companion visualization at `blauewelt.github.io/earth` renders most of these sources as interactive globe layers with per-dataset documentation; the descriptions here follow its catalog.)

GLORYS ocean reanalysis (CMEMS). An eddy-permitting model of the global ocean constrained by satellite altimetry, SST, sea ice, and in situ profiles — the closest thing oceanography has to a best-estimate movie of the ocean since 1993. Contributes monthly *surface current speed* ($\sqrt{u^2+v^2}$; the western boundary currents that carry the AMOC's upper limb are current-speed structures) and *mixed-layer depth* (log-transformed; deep winter mixing in the subpolar gyre is where surface waters acquire the density to sink — the "engine room" of the overturning). At 1° these come from the quarter-degree ensemble member binned down; the quarter-degree tensor reads the native grid and adds *sea-surface height*, whose meridional gradient is the geostrophic surface flow and whose boundary difference is the classical altimetry AMOC proxy [1].

Roemmich–Gilson Argo climatology. The gridded monthly temperature and salinity fields (1°, 0–2000 dbar) built from the Argo float array [21] — the backbone subsurface observing system: ∼4,000 autonomous floats, each profiling every ∼10 days. Contributes T and S at 4, 8, or 16 pressure levels (10–1900 dbar) depending on tensor generation. These are the density channels: thermal wind makes the transport an east-minus-west density difference, and the deep levels see the cold limb the surface channels cannot. Intrinsically 1°-smooth (in the quarter-degree tensor they are bilinearly upsampled and carry no sub-degree information — stated, not hidden). The record begins in 2004; earlier months are missing tokens.

NCEP/NCAR Reanalysis 1 surface wind stress. The longest-running atmospheric reanalysis [22], daily since 1948, on a Gaussian grid regridded to the tensor. Contributes monthly-mean τ_x, τ_y (Ekman transport, the wind-driven part of the AMOC's monthly variability — and hence also the source of the wind-only baseline every codec must beat) and the *within-month standard deviation* of the daily stress (storminess: buoyancy loss in storms drives convection in a way monthly means erase). Its length is what lets the quarter-degree tensor's axis start in 1982.

NOAA OISST v2.1. Daily 0.25° SST from AVHRR plus in situ data since late 1981 [23], monthly-meaned; optional channel 15 of the deep tensor, and (as a fixed 1991–2020 climatology) one of the two deliberately inert channels of §9.

GPCP v2.3 precipitation climatology. The satellite–gauge merged precipitation record [24]; enters only as the second inert climatology channel of §9.

Truth arrays (never inputs). Five basin-scale transport series, each a moored or cable observing system: *RAPID–MOCHA* 26.5°N [25], the flagship full-depth AMOC record (2004–, our 240 labelled months); the *Florida Current submarine-cable transport* [26], the lowest-latency AMOC component, daily since 1982 — the decade 1982–92 is usable truth only for the quarter-degree tensor's extended axis; *MOVE* 16°N [27], the longest deep-limb record (southward NADW flow, 2000–); *OSNAP* [28], the subpolar array where overturning variability originates (2014–); and *SAMBA* 34.5°S [29], the only sustained South Atlantic array (2009–, anomalies only).

3.2 Tensor generations, and why they may not be compared

The tensor is rebuilt from the fetcher scripts at the start of every training job, so a change to a fetcher silently changes the input of every subsequent run. Sampling 8 Argo pressure levels instead of 4 took the tensor from 14 channels to 24 (25 with monthly SST) without any dispatch asking for it. Every result below therefore belongs to one of three *tensor generations* — the **coarse tensor** (1°, 4 Argo levels), the **deep tensor** (1°, 8 levels), and the **quarter-degree tensor** (0.25° NA, 16 levels, SSH, axis from 1982) — with a fourth and fifth (the **pentad** and **daily** tensors of §3.3, same window and channels, finer time axes) in training as this revision is written, carrying no results yet — and cross-generation comparison is not licensed: the persistence baseline that chan% is measured against moves with the channel set, the wind-only ridge baseline moves with the grid (0.531 on the 1° tensors, 0.568 on the quarter-degree), and, as §7.5 shows, so does the meaning of the transport probe. (Earlier project notes call these "channel families" 1–3; run names keep the shorter prefixes.) Channel counts in this report are read from each checkpoint, never from a run's name — a run directory carried a misleading name for an hour before this was enforced, and every run now ships a provenance manifest recording the tensor it actually saw.

3.3 Input data

The subsections above are the history; this one is the specification, current as of the two cadence rebuilds of the quarter-degree tensor — **family 4** (pentad) and **family 5** (daily), which share family 3's grid and channel set exactly and differ only in the time axis. The encoder sees one array, $[T, 281, 481, 39]$: a 0.25° North Atlantic window (0–70°N, 100°W–20°E) whose samples sit *on* multiples of 0.25° rather than at cell centres, 39 channels, stored `float16` and z-scored per channel. The time axis is fixed bins counted from the epoch 1982-01-01 — monthly for family 3, 5-day for family 4 (T=3142), 1-day for family 5 (T=15706) — the same bin definition the transport labels are reduced onto, so a label is the target of a state with no re-alignment step anywhere. Table 2 is the channel list, in tensor order.

A second axis crosses the cadence axis: the *recipe revision.* Revision **r1** is the 39-channel tensor every number in this report was measured on. Revision **r2** appends one channel — daily 0.25° OISST sea-surface temperature, channel 40 — and is built, tested and published as an artifact, with no codec trained on it yet. It is listed here because the two revisions are one specification with one difference, and because a channel addition is not a variation of a tensor but a new one: it changes C, so it changes the encoder's input projection and channel embedding, and every codec must be retrained rather than fine-tuned. Appending rather than inserting is what keeps the two comparable — channels 1–39 hold the indices at which every published result was measured, and an r1 and an r2 tensor are diffable channel by channel.

Labels are not inputs. The transport series ride along as never-input targets, reduced to bin means on the same axis: RAPID 26.5°N overturning transport (240 monthly, 1,459 pentad, 7,290 daily labels) and the NOAA Florida Current submarine cable (daily since 1982; 433, 2,553, 13,931). The anomaly baseline used for training and probing is a *train-years-only* monthly climatology computed inside the pipeline — never the WMO 1991–2020 normal that the companion visualization uses, which contains held-out years and would leak test-period information into training through the baseline. The two baselines look interchangeable and are not.

The temperature channel, and what is still not an input. Every result in this report was measured on r1, and r1 carries no sea-surface-temperature field of any kind. Its only temperature is Argo `rg_t`, which begins in 2004 and is live one bin per month, so **22 of the 43 years on the axis carry no temperature at all**, and inside the Argo era the channel is a missing token in ≈83% of pentad and ≈97% of daily bins. That is the gap revision r2 closes, and it closes only that: daily 0.25° OISST is the sole channel native to the tensor's own grid and the sole channel live across the whole 1982–2024 axis, including the GLORYS-less 1982–92 decade in which the Florida Current cable supplies the labels. Because the source's grid is offset from the tensor's by exactly half a cell in each axis — OISST posts at cell centres $0.125° + k \cdot 0.25°$, the window samples *on* the multiples — the channel is bilinearly interpolated rather than nearest-sampled; nearest-indexing would displace the entire field by half a cell, consistently, and would look like an ordinary SST map. The planned test is a matched pair against the r1 codec in which the tensor is the only difference, so what SST is worth can be read off directly.

Three things remain deliberately absent. There is **no air–sea heat or freshwater flux**: the ocean's thermal forcing enters only through the state fields it has already acted on. There is **no sea-ice field**, although the window reaches 70°N. And there is **no CO_2 or radiative-forcing series**, which is a decision rather than an omission — over 1982–2024 monthly CO_2 correlates with a linear time index at $r = +0.99$, so a model handed it has been handed the calendar, and would fit the trend while being indistinguishable from one that learned the date.

ch.	channels	source	native	into the pentad tensor	into the daily tensor	live coverage
1–3	`cur_speed`, `log_mld`, `ssh` (3)	CMEMS GLORYS12V1 reanalysis (`uo`, `vo`, `mlotst`, `zos`)	1/12°, daily	5-day means of the daily fields, binned to the point-aligned 0.25° grid; `cur_speed` is the hypotenuse of the *binned mean components*, never a mean of magnitudes; `log_mld` is $\log_{10}$ of `mlotst`	the day's own fields	1993-01-01 to 2024-12-31; 1982–92 enters as missing tokens
4–35	`rg_t` (4–19) and `rg_s` (20–35), each at the same 16 pressures, 10–1900 dbar (32)	Roemmich–Gilson Argo climatology (Scripps): monthly mean + anomaly	1°, monthly	one live bin per month — the bin containing the 15th — lifted to 0.25° by NaN-aware bilinear; missing tokens in every other bin	the same policy, one live day per month	2004-01 onward
36–37	τ_x, τ_y (2)	NCEP/NCAR Reanalysis 1 momentum flux (`uflx`, `vflx`), sign-flipped so positive is stress *on* the ocean	≈ 1.9° Gaussian, daily	5-day mean	the day's field	the full axis, 1982–2024
38–39	τ_x std., τ_y std. (2)	as above	≈ 1.9° Gaussian, daily	within-pentad standard deviation	σ over the 5 calendar days *centred* on the bin's midpoint	the full axis, 1982–2024
40 (r2)	`sst` (1), *revision r2 only*	NOAA OISST v2.1 `sst.day.mean` (AVHRR analysis, NOAA PSL)	0.25°, daily	NaN-aware mean of the bin's 5 days, on the window's own grid	the day's own field	the full axis, 1982–2024; live in ≈100% of bins

Table 2: **What the encoder tensors contain**, families 4 and 5. Both are $[T, 281, 481, 39]$ `float16` at recipe revision r1 — $[T, 281, 481, 40]$ at r2, whose appended `sst` channel is the last row and is carried by no trained model yet — z-scored per channel, on the axis described above. The *ch.* column gives each group's 1-based position along the channel axis, in tensor order; that order is defined once, in `ml/build_family3.py`, and imported by every later builder, so the index a result quotes and the index the encoder reads cannot drift apart. The daily storminess channels are the same quantity as the pentad ones — for a pentad bin the centred 5-day window *is* the bin — sampled every day instead of every fifth, which is what keeps the two cadences comparable. The Argo policy is deliberate: forward-filling was rejected because it would tell the model the subsurface was observed on days it was not, so even inside the Argo era the one-live-bin rule leaves ≈ 83% of pentad bins and ≈ 97% of daily bins carrying the missing token on those 32 channels (92% and 98% over the whole 1982–2024 axis). Missingness is information, and it is represented, never imputed. The r2 channel is the complement of that row: OISST is observed daily on the tensor's own grid, so it is the one channel live in essentially every bin of the full axis, including the 22 years (1982–2003) in which the tensor otherwise holds no temperature at all.

Provenance. Every training run records the built tensor's sha256, shape and channel list beside the verbatim dispatch inputs, in a provenance manifest written after the build. A result therefore names the exact bytes it was measured on, rather than the recipe it was supposed to have used.

4 Methods

Stage 1 — the codec. A per-pixel masked autoencoder producing $z \in \mathbb{R}^{64}$ per pixel-month, trained on all pixel-months outside the holdout blocks. The pilot is 0.92M parameters; a 10.25M variant (320×8, the size the Chinchilla-style arithmetic of the Compute paragraph anchors for the current tensor) is training at time of writing. Checkpoints carry their architecture, and every loader reconstructs from it.

What exactly enters the encoder. The encoder is a small transformer (128×4 in the pilot, 320×8 in the 10M variant) over $C+2$ tokens for one pixel-month:

- **One token per channel**. If the channel is observed and not masked, the token is a linear projection of its scalar value (in anomaly space: the departure from that pixel's own monthly climatology, standardized) plus a learned per-channel identity embedding. If the channel is *masked for training*, the value is replaced by a learned `mask` token; if it is *genuinely unobserved* — Argo before 2004, polar gaps — by a distinct learned `missing` token. "I am hiding this from you" and "the observing system never measured this" are different facts, and the distinction measurably matters.
- **One context token**: a linear projection of $[\sin(2\pi m/12),\ \cos(2\pi m/12),\ \text{lat}/90,\ \text{lon}/180]$ — season and position. Never masked. This token is why static climatology input channels carry zero information (§9).
- **One CLS token**, whose final hidden state is projected to $z \in \mathbb{R}^{64}$.

The decoder never sees the inputs — it answers *queries* (channel, Δlon, Δlat, Δmonth) from z alone, with offsets up to ± 3 cells and months, so the training signal is reconstruction of masked channels *plus* prediction of spatial and temporal neighbours.

The patch codec. At 14 channels a pixel-month carries on average 10.4 observed values, so a 64-dimensional z is an *expander* ($\sim 6\times$), as in masked language models; the information bottleneck is the masking objective, not the dimensionality. The patch variant gives each channel token its 3×3 neighbourhood — nine values and nine observed-flags, projected by $\text{Linear}(2p^2, d_{\text{model}})$ — so that at 24 channels roughly 160 observed values enter one 64-dimensional vector, a genuine 2.5:1 compression. Longitude wraps (the globe is periodic in x) and latitude clamps with out-of-range rows marked unobserved; the centre cell defines whether a channel counts as observed, and reconstruction targets remain centre values, so the loss is unchanged and p=1 checkpoints load identically. The motivation is physical as well as informational: thermal wind *is* a density gradient, and a gradient does not exist within a single pixel.

Stage 2 — the temporal model. A causal transformer (default d_{model}=96 × 3 layers, 0.35M parameters; capacity sweep in Fig. 11) over each pixel's 24-month embedding sequence, predicting z_{t+1}; scored after decoding to channel space. The codec stays frozen.

Probes — the ladder. The base instrument is a ridge regression from section-mean embeddings to each truth series, deseasonalized, regularization chosen on a training tail — never on test data; verification is year-blocked k-fold with block-bootstrap confidence intervals. Two stronger read-outs sit above it, behind the same folds and the same inner-tail discipline. A small MLP ($64 \to 32 \to 1$, weight decay, early stopping, three seeds per fold) tests whether skill is present but not *linearly* accessible. An attention head tests whether skill is present but not accessible from the *pooled* section: every (pixel, month) embedding enters as a token with a longitude encoding, one learned query cross-attends over them, and a two-layer read-out maps the pooled vector to transport ($\sim$25k parameters, weight decay 10^{-2}, token dropout, three seeds per fold). The codec stays frozen throughout — these are probes, not fine-tuning. The design principle: each rung adds one capability, so a gap between rungs is attributable to that capability alone. Beneath the ladder sit two end-to-end *controls*: the identical head fed raw section values and observed-flags instead of embeddings, at single-pixel and at 3×3 receptive fields — pretraining is then the only manipulated variable (§7.3).

Rollout. For horizons $h = 1 \dots 12$: initialize with true embeddings, predict one month, feed the prediction back, repeat — scored in channel space against climatology, persistence, and damped persistence, and probed for the AMOC at each horizon.

Compute. Runs through 2026-08-07 were trained on GitHub-hosted 4-core CPU runners at ∼1.3 steps/s. Since then three rented consumer GPUs (RTX 4090, ∼$0.24/h each) are attached as self-hosted runners at ∼124 steps/s each — a factor of 95 per box, which turns a 30k-step schedule from six hours into four minutes and made the controlled comparisons of §7 affordable. The raw source data (∼13 GB) is mirrored as a GitHub release so a fresh box seeds its cache from a CDN rather than the origin archives, one of which (the Argo host) times out under load. Nothing about the model or protocol changed with the hardware. Chinchilla-style anchor for sizing: the current tensor carries ∼270M observed values per epoch; at the ∼20:1 data-to-params ratio that suggests a ∼13M-parameter codec — 14× the pilot, and the reason the 10M variant exists. The arithmetic is redone whenever the data changes.

5 Metrics and their statistical power

Full definitions are in the glossary; the reliability ordering is the substance. **chan%** is decision-grade within a channel set: seed spread ±0.5 points, so differences ≥ 2 points are real. The **fixed-holdout probe** r (36 held-out months, effective sample ≈ 15 after autocorrelation) has a 95% CI of ±0.57 — a compass, never a verdict; the same configuration has drawn 0.09 and 0.33 on two seeds. The **year-blocked k-fold** tests all 240 RAPID months once each (SE ≈ 0.10) and is the only probe number we argue from. The most instructive failure: a d_z=128 codec set the all-time fixed-holdout record (0.528) while the k-fold read 0.166 — the coarse instrument flatters widest exactly where overfitting is easiest.

The stage-2 programme of §8 adds two instruments at opposite ends of the power spectrum. The **forecast ratio** is the sharpest number in the project:

$$\text{ratio} = \frac{\text{MSE}\big(\hat{z}_{t+1},\, z_{t+1}\big)}{\text{MSE}\big(z_t,\, z_{t+1}\big)}$$

over held-out windows — the temporal model's one-step error in embedding space, divided by the error of simply asserting that next month's embedding equals this month's (persistence, whose measured MSE is ≈3.1 z-units^2). A ratio of 0.13 therefore reads "the model's one-step error is 13% of the no-change forecast's." Because every stage-2 arm shares the same frozen codec, the denominator is identical across arms and the ratio is directly comparable across the whole of §8 — which is exactly why it, and not a channel-space score, is the ladder's screening metric. Its statistical power is what makes the paired designs work: three-seed spread is ±0.002, so paired same-seed differences of 0.002 (the width effect) are readable and differences of 0.03 (the capacity effect) are $\sim 30\sigma$. Its limitation is inherited from its space: it is an internal, embedding-space quantity, blind by construction to anything the frozen codec cannot represent, and a gain on it need not survive iteration — which is what the **corridor AUC** exists to test. That instrument is the opposite trade: physically meaningful (channel-space MSSS against climatology, rolled twelve months, averaged over horizons, scored on the 29,627 corridor pixels), expensive (hours per head), and guarded rather than powerful — its error discipline is the gate head's exact reproduction (0.643, third decimal, every run) plus paired seeds, and its decision threshold is the observed seed spread (±0.005 at the 88M tier; at 205M, ±0.001 on the 60k-step triple and 0.002–0.005 across the five 200k-step pairs — the full replicate record, with the single-seed rule it licenses, is in the repository's experiment log). Every claim in §8 names which of the two instruments it is read from.

6 Related work

Three literatures border this project; we take them in turn. (The repository file `ml/LITERATURE.md` carries the full survey and is updated more often than this section.)

study	inputs	target & filtering	protocol	skill
Frajka-Williams [1]	altimetry + **cable** + Ekman	RAPID MOC, 18-mo low-pass	in-sample	$r \approx 0.95$
Sanchez-Franks [2]	altimetry (thermal wind)	RAPID, deseas. + 18-mo	in-sample	$r = 0.83$
Worthington [8]	boundary densities	RAPID *UMO*, monthly	1 test year	$R^2 = 0.78$; $r \approx 0.75$
Caesar [9]	subpolar SST fingerprint	AMOC index, 20-yr LOWESS	model-calibrated	trend only
Solodoch [3]	OBP+SSH+SST+τ	**ECCO** MOC, monthly	chrono. 70/30, *simulation*	$r \approx 0.98$
Meng [4]	SSH+OBP maps	eddying-model MOC	*simulation*	skill ≈ 0.44
Wölker [5]	virtual Argo + τ + cable	VIKING20X 26.5°N	*simulation*	$R^2 \approx 0.74$
this work (ridge)	14-ch embedding, **no transport**	**real** RAPID, monthly deseas.	year-blocked k-fold	$r = 0.602$ [0.46, 0.73]
this work (head)	24-ch embedding, **no transport**	**real** RAPID, monthly deseas.	year-blocked k-fold, 2 seeds	$r = 0.672$
same, under their filter	—	+18-mo low-pass	*as above*	$r = 0.33$–0.75
classical inputs re-scored in OUR protocol (§6.2) — same 227 months, same folds				
wind only	section τ_x, τ_y	real RAPID, monthly	year-blocked k-fold	$r = 0.401$ [0.27, 0.56]
cable only	**Florida Current transport**	real RAPID, monthly	year-blocked k-fold	$r = 0.254$ [0.15, 0.36]
cable + wind	**two of RAPID's three terms**	real RAPID, monthly	year-blocked k-fold	$r = 0.566$ [0.45, 0.68]

Table 3: **Joint results table.** Every row is a defensible number in its own frame; the frames differ so much that the rightmost column should not be read as a ranking. §6.1 explains each difference and quantifies the largest. Bold marks the two design choices that most change the task: whether a *transport measurement* is among the inputs, and whether the target is the *real* ocean or a simulation.

AMOC reconstruction from observables. The classical line regresses the RAPID transport on physically chosen observables. Frajka-Williams [1] reconstructs the MOC from satellite sea-level anomaly plus the cable and Ekman terms and explains >90% of variance ($r \approx 0.95$) — but after 18-month low-pass filtering, with in-sample calibration; Sanchez-Franks et al. [2] reach $r = 0.83$ from altimetry via a thermal-wind argument under the same filtering; Worthington et al. [8] fit boundary densities to the upper-mid-ocean transport ($R^2 = 0.78$ monthly, held-out $r \approx 0.75$ on a single 2017–18 test year). Fingerprint methods [9] recover multi-decadal AMOC trends from SST patterns but no monthly signal. The machine-learning line trains almost exclusively on simulations: Solodoch et al. [3] report $r \approx 0.98$ for a CNN reading OBP/SSH/SST/wind strips — inside the laminar 1° ECCO state estimate; in eddy-resolving models the same task drops to explained variance 0.44–0.74 [4, 5], which we read as the honest difficulty scale. Application to real RAPID data at monthly cadence with blocked cross-validation is, as far as we can determine, absent from the literature. Our pooled-ridge $r = 0.60$ — monthly, deseasonalized, real data, year-blocked k-fold, no RAPID-derived inputs — sits where the eddying simulations predict it should (the attention-head read-out reaches 0.65–0.69 across two seeds, with the attribution caveats of §7.3), and no published surface-to-OSNAP reconstruction exists for our OSNAP null to contradict.

6.1 Why these numbers are not comparable

Table 3 spans $r = 0.44$ to $r = 0.98$, and almost none of that spread is model quality. Four mechanisms, in descending order of how much they matter.

1. Some inputs already contain the answer. RAPID does not measure one thing; it sums three: the Florida Current transport, the wind-driven Ekman transport, and the upper-mid-ocean geostrophic transport. Frajka-Williams [1] takes the *cable* record (which *is* the Florida Current term, measured) and Ekman (from wind) as inputs, and reconstructs only the third. Two of the three addends are handed to the model; the reported correlation is substantially the correlation of a partial sum with its own total. Wölker [5] likewise includes cable transport. Our inputs contain no transport measurement of any kind — not the cable, not Ekman as a transport — so we are inferring all three terms from the ocean state. These are different problems, and the harder one should be expected to score lower.

2. The 18-month filter removes 83% of the variance and 87% of the degrees of freedom. This is the difference most often glossed as a methodological detail, so we measured it on the RAPID series itself. Monthly deseasonalized RAPID has $\sigma = 2.79\,\mathrm{Sv}$, a lag-1 autocorrelation of 0.32, and an integral timescale of 3.5 months — so its 240 months carry $\approx$68 effective degrees of freedom. After an 18-month low-pass, the integral timescale becomes 25 months and the effective sample size falls to $\approx$**9**, while the retained variance is **17%** of the original. Two consequences. First, the filtered task is easier: what survives is the slow, heat-content-like component that altimetry and density observables track well, while the wind-driven month-to-month variance — the part we are scored on — has been removed. Second, the filtered statistic is barely constrained: $r = 0.95$ on 9 effective points carries a 95% confidence interval of roughly $[0.77, 0.99]$, and $r = 0.83$ one of $[0.37, 0.96]$ — intervals wide enough to overlap our unfiltered number. (As a check on the arithmetic: the same calculation gives $[0.42, 0.73]$ for our $r = 0.60$ at 68 effective points, against the $[0.46, 0.73]$ our block bootstrap reports independently.)

We can also show the instability empirically rather than asserting it. Two codecs of *identical* configuration differing only in training seed score 0.543 and 0.531 monthly — a gap of 0.012 — but 0.449 and 0.327 once 18-month filtered, a gap of 0.122, ten times larger. Across our runs the filtered correlation ranges from 0.33 to 0.75 while the monthly one stays within 0.53–0.62. We report the filtered number for every run (it is in the repository's leaderboard), but we decline to headline a statistic whose seed noise is an order of magnitude larger than the effect anyone wants to measure.

3. In-sample calibration is a fit, not a prediction. The classical reconstructions [1, 2] tune their coefficients on the same period over which skill is then reported; Caesar [9] calibrates against models. No held-out set exists, so the number answers "how well can this functional form describe these years" rather than "what would it say about a year it has not seen". Worthington [8] is the honourable exception in the classical line, holding out 2017–18 — one test year, where our protocol re-fits from scratch for every year in turn. Our year-blocked k-fold is the strictest arrangement here and costs us skill relative to every in-sample row, by construction.

4. A simulated ocean is a different ocean. Solodoch's $r \approx 0.98$ [3] is obtained inside ECCO, a 1° state estimate whose dynamics are laminar and whose surface fields and MOC are generated by the same model — the mapping to be learned is close to deterministic and free of observational noise. The moment the same task is posed in eddy-resolving simulations it falls to 0.44–0.74 [4, 5], and the real ocean adds sparse, irregular sampling on top of eddies. We read the eddying-model range, not the laminar one, as the relevant difficulty scale — and our 0.60–0.67 on real observations sits inside it.

6.2 Making them comparable: the classical inputs under our protocol

A like-for-like comparison requires real RAPID data, monthly resolution without low-pass filtering, no transport measurement among the inputs, and an out-of-sample protocol; no published result meets all four. But the comparison can be rescued from the other end. Rather than dress our number up in the literature's conventions, we put the literature's *inputs* through our protocol — same target, same deseasonalization, same year-blocked folds, same block bootstrap — so that the rows differ only in what the model is allowed to see. The Florida Current cable, normally one of our never-input truth series, is used as an input for these rows and these rows only (`ml/classical_baseline.py`).

The bottom block of Table 3 reports it, on the 227 months where RAPID and the cable overlap. Wind stress alone gives $r = 0.401$; the cable alone $r = 0.254$; **the two together — i.e. two of RAPID's three constituent terms, handed to the model as direct measurements — give** $r = 0.566$ [0.45, 0.68]. Our embedding, which is shown no transport measurement of any kind, reaches 0.578–0.620 under the same pooled ridge and 0.672 with the unpooled head.

We read that carefully. It does *not* say we beat Frajka-Williams: that reconstruction also uses altimetry for the third term, which this ladder omits, and it targets a filtered series. What it does say is sharper and is the number we could not get any other way: at monthly resolution, out of sample, being handed the Florida Current and the Ekman transport outright is worth about $r = 0.57$ — and a task-blind embedding of the ocean state matches it without being handed anything. The gap between that 0.57 and a published 0.95 is therefore not model quality; it is the 18-month filter and the in-sample fit, quantified. A supporting observation in the same direction: when we apply the 18-month filter to these classical rows' *out-of-fold* predictions their correlations collapse (-0.07, -0.23, $+0.24$), which is what a filtered statistic with nine effective points does once it is no longer allowed to see its own test years.

Ocean and climate forecasting. Our field-prediction metrics are the standard decadal-verification quantities under other names: skill vs. climatology is MSSS with a climatological reference, and we adopt ACC per lead and damped persistence from the same framework [6]. Reference points at our monthly cadence: Ham et al.'s ENSO CNN [10] holds Niño3.4 correlation > 0.5 to $\sim$17 months (the ocean's most predictable index); Taylor & Feng [11] forecast global monthly SST anomaly fields with a Unet–LSTM (RMSE 0.48°C at 1 month rising to 0.63 at 18) — the closest published analogue to our stage-2 rollouts. The daily 1/4°–1/12° neural ocean forecasters (XiHe, WenHai, GLONET) and the global weather models they descend from [12, 13] operate in a different regime (days, full fields, reanalysis-supervised) and are cited for orientation, not comparison.

Self-supervised and foundation models for Earth observation. Architecturally we are a masked autoencoder [14] over channel tokens rather than image patches. The flagship Earth-system foundation models — ClimaX [15], Aurora [16], and the land-surface line of Prithvi [17] and SatMAE [18] — evaluate mainly by full fine-tuning; frozen-representation evaluation, our entire protocol, is the exception: Prithvi WxC's anomaly-space masked pretraining with frozen backbone and small heads is our closest architectural relative (atmosphere-only), and Lehmann et al. [7] argue explicitly that frozen-backbone extension should be a standard foundation-model quality metric. AlphaEarth Foundations [19] embeds every terrestrial 10 m pixel into 64 dimensions — the same width we use — from ten data modalities, and is the clearest statement of the "embedding field" idea this project transplants to the ocean; it is terrestrial-only, and its modality list (translated to ocean: altimetry, gravimetry, scatterometry, ocean colour, sea ice) is our data-ladder shopping list. Model sizing follows the compute-optimal-scaling heuristic of Hoffmann et al. [20] adapted to observed-value counts (§4, Compute). To our knowledge no self-supervised foundation model exists for the gridded ocean interior; frozen-embedding probes to integrated circulation indices, and missingness-as-information over the historical observing system, remain unoccupied territory.

7 Results

7.1 The bottleneck: what width buys

Fig. 1: the transport read-out wants more width than field reconstruction rewards, but not unboundedly — the turnover at d_z=128 is a truth-data limit, not a representation limit. That this sweep was run at C=12 becomes important in §7.5.

7.2 The probe ranking, and what the wind already knows

Fig. 2 is the project's headline and its conscience in one picture. The wind channels moved the probe more than every architecture decision combined ($0.31 \to 0.60$); the global codec matches the regional one exactly (0.602 vs 0.604 — the generic-embedding hypothesis holds at fixed capacity); and the wind-only baseline at 0.53 says most monthly skill is Ekman physics. The coarse tensor's embedding increment over pure wind is $+0.07$; its distinctive contributions are the low-frequency component (18-month low-passed $r = 0.64$) and event anatomy (Fig. 6). In Sverdrups: RMSE 2.23 Sv against a target standard deviation of 2.79 Sv.

run	window	C	patch	d_z	steps	chan%†	k-fold r [95% CI]	dip
coarse tensor (1°, 4 Argo levels)								
pilot4	NA	4	1	32	8k	+29.3	—	—
dz8	NA	12	1	8	30k	+28.6	0.111 [0.01, 0.20]	—
dz16	NA	12	1	16	30k	+30.3	0.151 [0.01, 0.28]	—
dz32	NA	12	1	32	30k	+30.5	0.182 [0.05, 0.31]	—
dz64	NA	12	1	64	30k	+30.6	0.308 [0.13, 0.46]	16%
dz128	NA	12	1	128	30k	+30.2	0.166 [0.07, 0.30]	—
wind14	NA	14	1	64	30k	+35.6	**0.604** [0.47, 0.72]	50%
global14	global	14	1	64	30k	+30.6	**0.602** [0.46, 0.73]	50%
global14b*	global	14	1	64	30k	+30.9	0.556 [0.43, 0.68]	—
global15sst	global	15	1	64	30k	+30.8	0.582 [0.43, 0.71]	47%
deep tensor (1°, 8 Argo levels; §3.2)								
patch24	global	24	**3**	64	40k	+32.0‡	0.543 [0.43, 0.66]	27%
pixel25	global	25	1	64	40k	+30.9‡	0.536 [0.38, 0.68]	**59%**
pixel24	global	24	1	64	40k	—	0.515 [0.41, 0.63]	—
pixel24-30k	global	24	1	64	30k	—	0.585 [0.45, 0.71]	—
pixel24-60k	global	24	1	64	60k	—	0.549 [0.42, 0.67]	—
patch24-30k	global	24	3	64	30k	—	0.551 [0.42, 0.68]	—
patch24-1M	global	24	3	64	**1M**	+32.1‡	0.571 [0.42, 0.70]	—
patch24-seed2	global	24	3	64	40k	—	0.531 [0.40, 0.65]	27%
pixel24-seed2	global	24	1	64	40k	—	0.564 [0.44, 0.69]	—
big10M (10.26M)	global	24	3	64	60k	—	**0.578** [0.45, 0.70]	26%
quarter-degree tensor (0.25° NA, 16 Argo levels, SSH; §7.4)								
f3-pilot	NA 0.25°	39	3	64	40k	—	**0.620** [0.48, 0.74]§	46%
f3-anchor41M (40.7M)	NA 0.25°	39	3	64	60k	+31.5	0.631 [0.51, 0.73]§	**51%**

Table 4: **Master table: every codec run.** Generated from each run's own JSON by `ml/make_table.py`, not transcribed. †chan% is comparable only within one channel set (its persistence baseline moves with the channels); cross-run comparison goes through the year-blocked k-fold RAPID correlation. Dip = share of the winter 2009–10 collapse amplitude captured out-of-fold. The wind-stress-only ridge baseline (no embedding) scores $r = 0.531$ — the line every codec must beat *on the* 1° *tensors*; the same baseline re-measured on the quarter-degree tensor is 0.568, which is what the § row must be read against — never against 0.531. k-fold r here is the POOLED RIDGE rung of the probe ladder (the comparable-across-time instrument); the unpooled attention-head values (§7.3) are higher for every codec measured: global14 0.635, pixel25 0.617, patch24 **0.690** and 0.654 on its second seed, and f3-anchor41M **0.662** [0.56, 0.75] against 0.628 [0.51, 0.73] for the matched raw-3×3 head on the same section — a pretraining margin of +0.034 from one seed, reported here only because the pooled row above it would otherwise be read as the codec's ceiling. *global14b is an exact-configuration replication on a different runner: it measures codec-training seed noise on the k-fold at $\approx \pm 0.05$ — the four pixel24/patch24 rows, spanning 0.515–0.585 across 30k–60k steps, are consistent with that noise and show no step-count trend; the patch24-1M row settles the saturation question directly — every probe metric is flat from the first 50k-step probe to the millionth step (§10), so the pilot size is capacity-limited, not compute-limited. ‡in-training probe at the final step, not the post-hoc trainprobe used for coarse-tensor rows. Stage-2 figures for global14b and global15sst are *withdrawn*, not stale: both were trained on a poisoned embedding cache (§10).

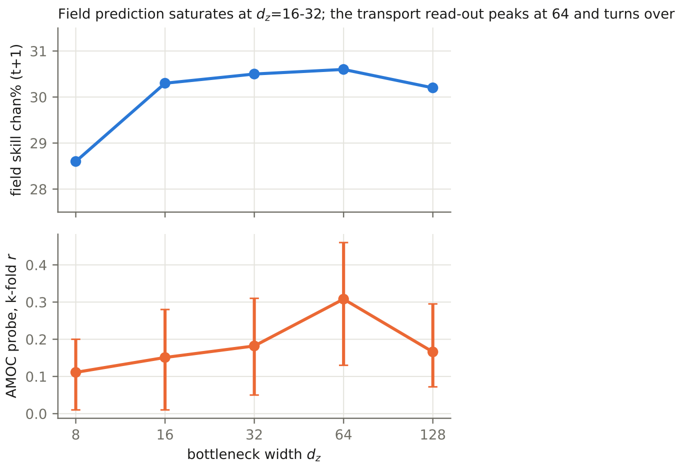

Figure 1: Bottleneck width d_z versus its two customers, on the 12-channel North-Atlantic tensor at matched training steps. Field prediction (top) saturates by d_z=16–32; the AMOC probe (bottom, with 95% block-bootstrap CIs) rises to a peak at d_z=64 and turns over at 128 — 128 ridge features against ~220 truth months per fold overfits. d_z=64 became the working setting, a choice made when the tensor had 12 channels.

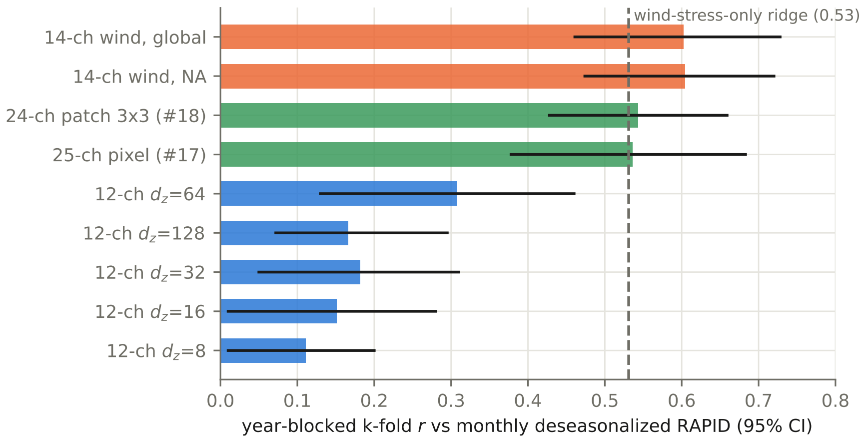

Figure 2: Year-blocked k-fold correlation with monthly deseasonalized RAPID transport, all codec generations, 95% CIs. Adding wind stress (orange) nearly doubled the defensible probe; training globally cost the Atlantic read-out nothing. The deep-tensor codecs (green) fall back onto the dashed wind-only baseline — a ridge on section-mean wind stress with no embedding at all.

7.3 The probe ladder: pooling was hiding the signal

Fig. 3 is this revision's central result, and it changes both a number and a story. The number: the patch codec read by the attention head reaches $r = 0.690$ [0.570, 0.781], RMSE 2.02 Sv (and 0.654 on a second codec seed, below) — against 0.60 for the best pooled-linear configuration, and against a wind-only baseline of 0.531 that the pooled deep-tensor probes had barely cleared. The story: the ladder's two rungs dissociate cleanly. Pointwise nonlinearity (MLP rung) recovers nothing on any target for any codec — so the ridge was never the limit as a *function class*. Spatial structure (head rung) recovers a great deal — so the limit was the *pooling*. Mean-pooling the section computes the one statistic in which an east-west difference cancels, and thermal wind says the transport *is* an east-west difference. The two architectural changes of this cycle turn out to be halves of one physical idea: the 3×3 patch puts local gradients into each embedding, the attention head reads gradients across the section, and each looked worthless without the other — the patch codec's pooled numbers were indistinguishable from its pixel control.

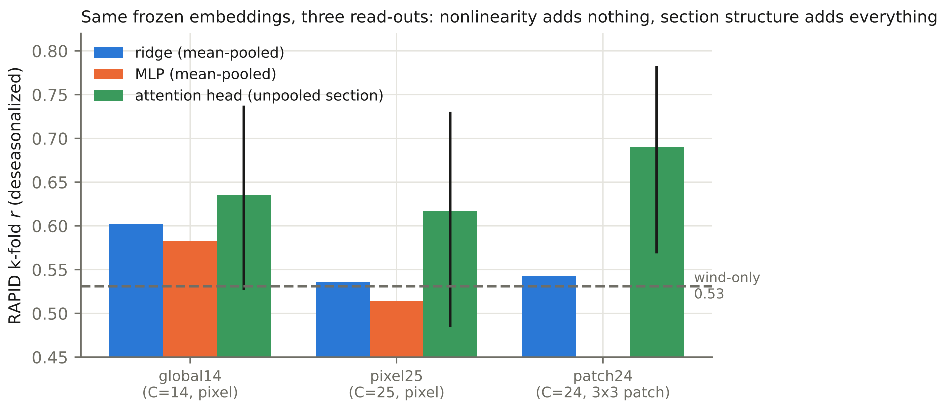

Figure 3: Same frozen embeddings, three read-outs, same year-blocked folds. The MLP (orange) trails the ridge (blue) everywhere — pointwise nonlinearity recovers nothing. The attention head over the unpooled section (green, with 95% CIs) beats every pooled probe, restores the 25-channel codec to parity with the 14-channel one, and reveals the patch codec — invisible to pooled probes — as the strongest representation in the programme at $r = 0.690$.

The attribution matrix. An end-to-end baseline completes the picture: the *same* attention head, same folds, same regularization, fed raw section anomalies (values + observed-flags) instead of embeddings, at both receptive fields:

tokens carry	raw data	codec embedding
single pixel	0.613 [0.48, 0.73]	0.617 [0.49, 0.73]
3×3 neighbourhood	0.659 [0.55, 0.75]	0.690 / 0.654 (2 seeds)

A second seed, and a retraction. The 0.690 cell was replicated on an independently trained codec of identical configuration. It came back at 0.654 [0.543, 0.748]. The capability claim survives — both seeds beat every pooled probe and the wind-only baseline decisively, so unpooling is real — but the *pretraining margin does not.* The raw-3×3 baseline is codec-independent at 0.659; the two-seed embedding mean is 0.672; the advantage is therefore +0.013, which is nothing. We previously reported +0.031 "positive in direction, not significant" and we withdraw even that directional reading: it was one seed's noise. Note that the pooled ridge replicated far more tightly across the same two codecs (0.543 vs 0.531) than the head did (0.690 vs 0.654) — the more expressive probe is also the noisier instrument, and no future head number in this programme should be quoted from a single seed.

What remains, then, decomposes as: *unpooling* the section is the whole of the improvement (pooled ridge 0.54 → head 0.65+); the 3×3 *receptive field* adds ∼+0.045 and does so on raw data alone; *pretraining* adds nothing measurable at either receptive field. For the monthly RAPID probe, a supervised head over raw section data recovers everything the codec offers. The pretraining case now rests entirely on the scoreboards where 240 labelled months have no answer: field prediction (stage 2 beats persistence everywhere; the matching end-to-end field-forecaster baseline is the open follow-up), transfer to sparsely-labelled targets like MOVE, and whether a margin appears at all once the codec is sized to its Chinchilla anchor.

That last test has now run, on the quarter-degree tensor at 40.7M parameters (§7.4), and it neither rescues the pretraining case nor closes it: the head reads 0.662 [0.557, 0.745] from the codec against 0.628 [0.514, 0.729] from raw 3×3 tokens on the same section, a margin of +0.034 where the 1° pair gave +0.013. Same sign, roughly twice the size, still a small fraction of its own confidence interval, and — the part that governs how it should be quoted — *one seed.* We wrote above that no head number in this programme should be quoted from a single seed, and we decline to exempt this one because it happens to point the way we would prefer.

At 5-day cadence the decomposition sharpens, and the representation term goes to zero. The quarter-degree tensor rebuilt at pentad cadence (T=3142, n=1,459 transport bins) supports the same comparison at six times the label count, and every rung of it has now been read under one protocol.

The pooled ridge *under-reads*: 0.660 [0.593, 0.722] against a wind-only bar of 0.670 measured on that tensor — on the pooled number alone this would have been recorded as another failure to clear wind. An attention head over the unpooled section pixels clears the bar: the frozen monthly codec reads 0.691 [0.631, 0.746], recovering the +0.031 that pooling had destroyed. But the matched raw-pixel control — the same head, same folds, fed section anomalies with no codec at all — reads 0.683 [0.620, 0.742]; and a codec trained from scratch *on the pentad tensor itself* (38M parameters, 167k steps) reads 0.680 [0.617, 0.740] through the identical head, −0.003 *below* its own raw control. Three read-outs that see unpooled section pixels — one frozen representation, one purpose-trained representation, and none at all — land within 0.011 of one another with nearly coincident intervals. So at 5-day cadence neither a frozen monthly representation nor 167k steps of matched-cadence pretraining adds measurable transport information beyond the raw fields: *read-out design, not representation learning, is what moved the number.* This also disposes of the domain-shift reading the pentad arm was built on — if monthly embeddings under-performed because 5-day fields are out of domain for them, a codec trained on 5-day fields should have beaten them, and it does not. Each of the three cells is one seed, and by our own rule we do not quote any of them as a level; the claim rests on their *differences*, which are 3–12× smaller than the head probe's own measured seed spread (0.036) and are therefore consistent with zero however the seeds fell.

The scope of that statement. It is a statement about the *current-state* read-out: what today's fields, encoded or not, say about today's transport. The codec's function in this system is not to add information — it is a compression and cannot — but to make the state *attendable*: to turn the 84,405 ocean pixels × 39 channels of one time step into a token sequence a transformer can carry and roll forward. The representation's operative test is therefore forecast skill from rolled-forward embeddings, the corridor AUCs of §8, and on that scoreboard the embedding pipeline is the only entrant: the matched raw-pixel control that makes the table above interpretable has no forecaster counterpart at this resolution, where the daily tensor alone is 165.6 GB and attention over raw fields is not tractable on any hardware we run. We report the parity and do not read it as a verdict on the representation.

Honesty box: the head is the most expressive probe we run (∼25k parameters against ∼220 training months per fold), and although it is regularized hard, early-stopped on a train tail, and seed-averaged behind the same blocked folds as every other number, the head-vs-ridge gaps have not yet been given a paired significance test. The second-seed replication is no longer outstanding — it is reported above, and it cost us the pretraining margin. A capacity sweep of the head itself (23k→325k→2.0M parameters) shows *neither* side is parameter-limited: at 325k both columns drop (raw-3 × 3 0.659 → 0.590, embedding-3 × 3 0.690 → 0.611), as the label-count arithmetic predicts (∼220 training months per fold support a head of tens of thousands of parameters, not hundreds), and the embedding margin persists at both sizes. The 2.0M cells are running as a formality.

7.4 First light from the quarter-degree tensor

The first codec trained on the quarter-degree tensor — a pilot-size (0.92M) control, C=39, 84,405 pixels, single seed — posts the highest pooled-ridge k-fold of the programme: RAPID $r = 0.620$ [0.484, 0.741], RMSE 2.25 Sv, 18-month low-passed 0.747, 46% of the 2009–10 dip. Three readings, in decreasing confidence. First, the baseline discipline holds: the wind-only ridge measured *on this tensor* is 0.568, so the codec clears its own baseline by +0.052 where the deep-tensor pilots sat exactly on theirs — though the CIs overlap and this is one seed. Second, MOVE transfers again (0.235 [0.112, 0.340], low-passed 0.726) while wind-only at MOVE is *negative* (−0.376): at 16°N the embedding carries signal the wind actively gets wrong. Third, and to be read with care: 0.620 beats every 1° codec on the same truth and protocol, but across tensor generations that sentence means "this data recipe is better", not "this codec is better" (§3.2).

The 41M-parameter codec trained *at* this tensor's compute-optimal anchor has since finished, and its answer is a flat one: pooled k-fold RAPID $r = 0.631$ [0.513, 0.732], RMSE 2.16 Sv, against the pilot's 0.620 on the same tensor and protocol. Forty-four times the parameters bought +0.011, comfortably inside seed noise. Both clear the quarter-degree wind bar (0.568) and neither does so significantly. The honest reading is that skill on this tensor is *not* capacity-limited at the pooled read-out — which, combined with the flat 50k→1M step sweep of §10, closes both easy scaling axes at once: neither more parameters nor more steps is what stands between this programme and a better transport number. What did move: the unpooled head reads 0.662 [0.557, 0.745] from the same frozen codec, the 2009–

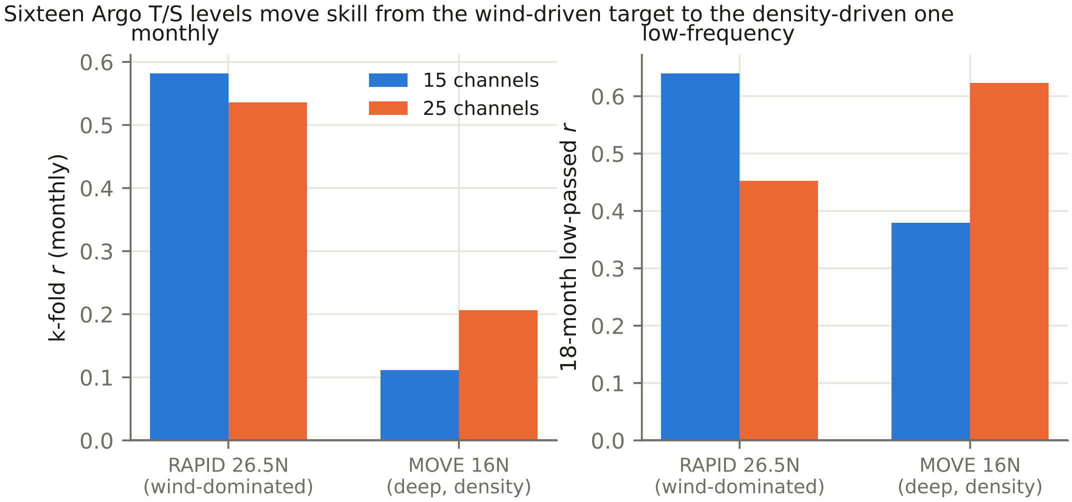

Figure 4: Sixteen Argo temperature and salinity levels to 1900 dbar *appeared* to cost monthly RAPID skill under pooled probes (left, first pair) — resolved by §7.3 as a read-out artifact, not a representation change — while genuinely roughly doubling skill at MOVE (left, second pair) and transforming the low-frequency read-out there (right). MOVE measures the *deep* western-boundary flow at 16°N — precisely the quantity the added levels describe.

10 dip capture rose to **51**% from the pilot's 46%, and the 18-month low-passed correlation reached 0.820 — the last of which should be read against the ~9 effective degrees of freedom of §5, not as a 0.82-with-240-months would be.

7.5 Where more data moved the skill

Under *pooled* probes, expanding the input appeared to cost RAPID skill: both deep-tensor codecs landed on the wind-only line (0.543 and 0.536 against 0.531) where the coarse tensor held +0.07. Section 7.3 resolves this: the skill had moved into cross-pixel structure, not out of the representation, and the unpooled read-out recovers it in full (0.617–0.690). The RAPID half of this subsection's original claim is therefore retracted; what follows — the MOVE result — is the part that survives, and it was measured pooled, so unpooling can only strengthen it.

At MOVE the same channels do the opposite. The 25-channel codec scores $r = 0.206$ [0.044, 0.346] — the first multi-target result in this programme whose confidence interval excludes zero — with an 18-month low-passed correlation of 0.623, against 0.111 and 0.379 for the 15-channel codec. MOVE is a deep, density-driven transport; RAPID's monthly variance is largely Ekman. The added channels are Argo T and S to 1900 dbar. The skill went where the physics says the information is.

The corrected reading is about read-outs, not bottlenecks: the embedding kept everything, and the probe determined what could be seen. For the generic-embedding hypothesis of §6 this is a *stronger* confirmation than the regional-versus-global comparison — one task-blind representation now serves the wind-dominated and the density-dominated target simultaneously, provided the read-out respects the structure of the question being asked of it.

One more asymmetry is worth recording: the 25-channel codec captures **59%** of the winter 2009–10 collapse out-of-fold, the best of any run in the programme, while its monthly correlation is the second worst. Event anatomy and average correlation are not the same capability, and a table that reports only the latter will hide the former.

7.6 How far ahead the ocean can be predicted

Fig. 5: feeding predictions back month after month, the anomaly field stays predictable beyond a year (MSSS +0.11 at h=12; ACC 0.36). The decay is decorrelation, not amplitude damping (the forecast-to-observed amplitude ratio holds near 0.5 throughout). The AMOC read-out on rolled embeddings, by contrast, dies after about a season — transport forecasting at range remains open, consistent with its

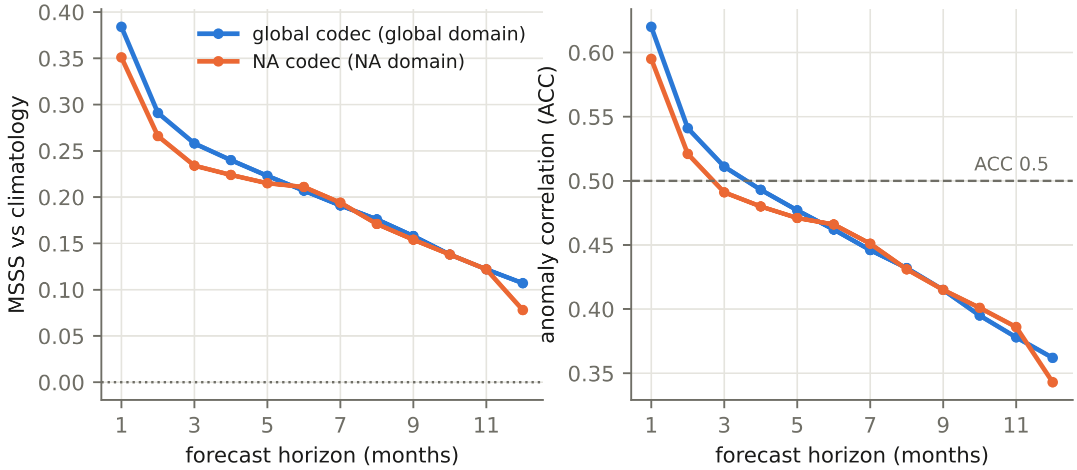

Figure 5: Autoregressive rollout skill versus forecast horizon for the global codec (blue, global domain) and the regional codec (orange, North-Atlantic domain; domains differ, so compare shapes more than levels). Left: MSSS against climatology — still positive at 12 months. Right: anomaly correlation, crossing the conventional 0.5 useful-skill line near four months.

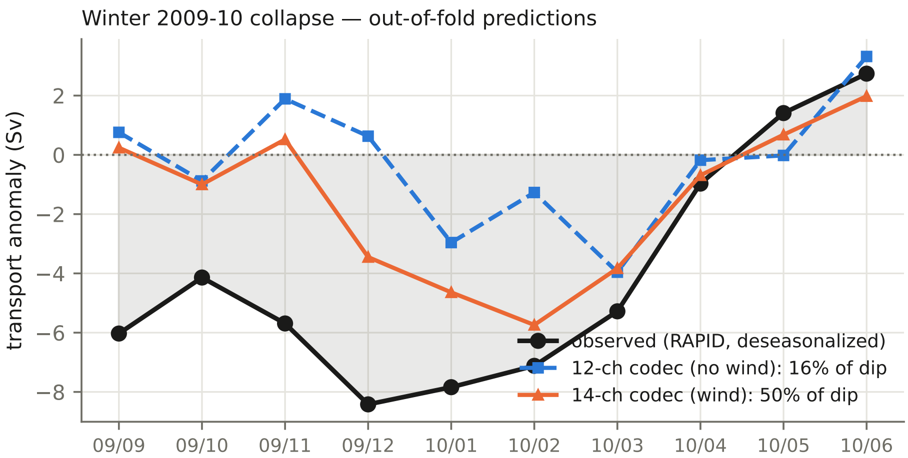

Figure 6: The sharpest event in the RAPID record, predicted out-of-fold. Without wind channels (blue) the model captures 16% of the dip amplitude; with them (orange), 50%. The 25-channel codec reaches 59% (not plotted; see Table 4). The November onset — triggered inside a month — stays invisible to monthly data.

wind-driven monthly variance.

7.7 Case study: the winter 2009–10 collapse

8 The rolled forecast at scale

Everything above probes what the representation *contains*. This section — the work of 13–16 August, and the reason for v6 — is about what the temporal model can *do* with it: predict the ocean's state forward, month over month, over the pixels that carry the overturning. One sentence of configuration so the reader carries the right picture: *everything in this section runs on the quarter-degree tensor* — all 39 channels of Table 1, including the sixteen Argo depth levels to 1900 dbar and sea-surface height — through the frozen 40.7M anchor codec of §7.4; the 14- and 24–25-channel tensors of the earlier sections are history here, and "scale" below always means the size of the *stage-2 temporal head*, never the codec or the channel count. The section's one-sentence summary: input geometry looked decisive and was a symptom; **capacity was the variable**, and it has not stopped paying at 205M parameters.

8.1 Spatial coupling: the stencil

The stage-2 model of §4 predicts a pixel's next embedding from that pixel's own history — an ocean of independent columns. Physically that is wrong in a specific way: anomalies *advect*, and a month of Gulf Stream transport moves water hundreds of kilometres. The stencil gives the temporal model that information: alongside the centre pixel's 24-month embedding sequence it reads the sequences of S neighbour pixels at fixed geographic offsets. Three geometry families were swept (rings at one radius; spirals, every point on its own bearing; and golden-angle *sunflower* layouts with a far-heavy $\sqrt{\cdot}$ radial ramp at Fibonacci counts), with reach up to 4444 km. Reach beyond 4444 km was measured and *refused* before any training: on this window it deletes live input (26.1 live corridor slots at $r_{\max}$=4444 km for a 55-point sunflower, 19.7 at 6666 km), because the extra reach lands on land or off-window.

Two scoreboards, defined in the glossary, are kept strictly apart — the **forecast ratio** (one step ahead, against persistence, reproducible to ±0.002 across seeds) and the **corridor AUC** (twelve months rolled, feeding predictions back, scored where the AMOC lives). The programme's founding observation (E-022, 13 August) was that they can *disagree*: a change that improves the one-step forecast can worsen what compounds under iteration. Every evaluation runs behind the **gate head** discipline — a frozen no-neighbour reference must reproduce its pinned corridor AUC of 0.643 exactly before any new number is accepted; it has now done so on every evaluation in this section.

A picture of spatial generalisation. Every rolled number in this section is a scalar summary of a field, and the field is not flat. Fig. 7 draws it — per-pixel skill against climatology at h=6 months over all 84,405 window pixels, for one seed of the widest 205M arm. Almost everywhere the basin beats climatology comfortably: 0.86 averaged over the trained longitudes, 0.86 again in a Gulf Stream box (32–45°N, 75–45°W), and only 2% of window pixels below zero. The exception is a single meridional band — and the band is not an ocean feature. Its edges fall at exactly 45°W and 25°W, which are the two numbers in `train.py`'s `--holdout-lon` default. That block is withheld from training in *both* stages: the stage-1 codec pool is built as `obs_any & ~t_hold & ~x_hold` and the stage-2 head pool as `ok_p = ~x_hold[xs]`. The rolled evaluator recomputes the identical mask from the checkpoint's own arguments, uses it to set the normalisation constants — and then scores every pixel anyway. For every evaluation reported here it recorded nothing about the block in its output at all; the evaluator has since been changed to emit the bounds, the per-scope in-block counts, and each scope split into its trained and held-out longitudes, so future rolls report the two separately. The band is therefore the model's *spatial generalisation gap*, and this figure is a direct picture of one: skill against climatology as a function of distance from the training set.

Three measurements identify it and close the alternatives. *(i) It is a boundary effect, not a patch.* Skill decays monotonically with distance *into* the block from each trained edge and does so from both sides: reading in from 45°W at one-degree steps, 0.70, 0.49, 0.34, 0.23, 0.19; in from 25°W, 0.71, 0.54, 0.42, 0.33, 0.26; the two meeting at ≈ 0.17 in the middle rather than collapsing to nothing. The ten steepest column-to-column gradients anywhere in the 481-column window all lie inside the block and within 2.5° of an edge; the figure's code asserts that rather than asking the reader to see it. The edges are ramps 1–2° wide, not discontinuities. *(ii) The mechanism is explicit.* The head's static context concatenates the pixel's normalised coordinates ($\varphi/90$, $\lambda/180$), so *longitude is a literal input feature with a 20° hole in its training range*: the head is interpolating across a gap in its own input, and its accuracy decays with how far into the gap it has to reach. *(iii) The alternatives fail.* It is not the codec — the retrained deep-temperature decoder reads 1.43% rmse^2 on the held-out longitudes against 0.85% on trained ones (and 1.90% on held-out *months*), so the representation generalises across the block essentially perfectly and the collapse is entirely a stage-2 forecast-head failure. It is not latitude-structured, which is what a Mid-Atlantic Ridge or an observing-coverage reading would require: in ten-degree bands the in-block/out-of-block contrast is 0.29/0.83 at 0–10°N and 0.42/0.91 at 60–70°N. And it is not the variance normalisation, which is the reading the previous revision of this paragraph offered. MSSS is a variance-normalised ratio, so we computed its denominator per pixel — the standardised anomaly variance of the 0.25° base tensor's three dynamic channels, on the evaluation's own recipe of train-month climatology with the holdout years excluded. It is smooth and edgeless straight through the region, with nothing happening at 45°W or 25°W; it is nearly *twice* as large at the western edge (1.38) as at the eastern one (0.79), where skill is the same to within 0.01; and it reaches its minimum

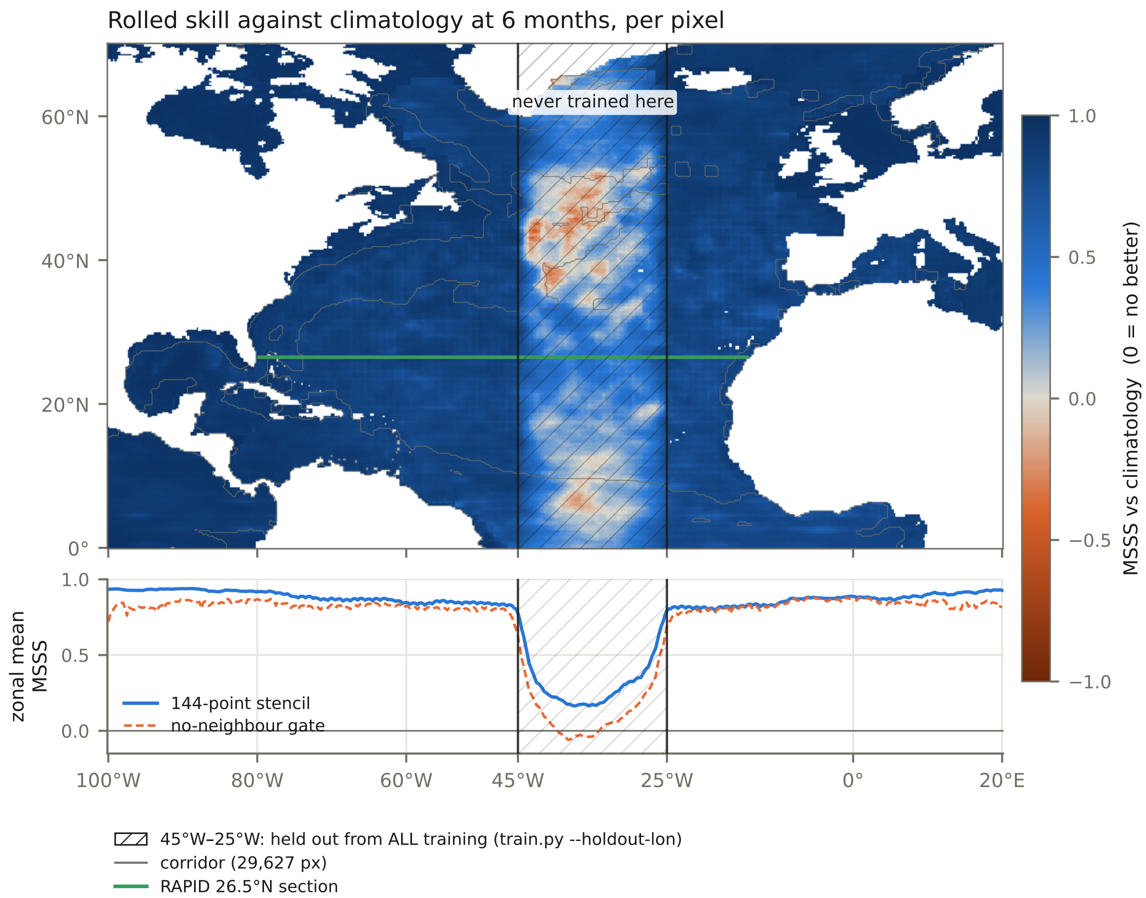

Figure 7: **A held-out longitude block, seen from outside it.** Per-pixel MSSS against climatology at a six-month horizon — pooled over all 39 channels and every evaluation start-month — for all 84,405 window pixels, from one seed of the 205M 144-point stencil head. Positive means beating the train-month climatology, zero means matching it, and the colour scale is symmetric about zero, so the midpoint is the climatological line itself. Land and out-of-window cells are blank: an unmodelled cell and a zero-skill cell are not the same thing. **Hatched: 45°W–25°W, the longitude block that `train.py --holdout-lon` withholds from training in both stages** — no codec fit and no forecast-training window ever falls there, while the rolled evaluation scores those pixels along with the rest. It is a quarter of the window. Grey: the 29,627-pixel corridor, whose outline follows the coast almost everywhere because the fastest quarter of the window largely is coastal. Green: the 265-pixel RAPID 26.5°N section (80°W–13°W), where the transport read-out is taken. *Lower panel*: the same field as a zonal mean, with the frozen no-neighbour gate head beneath it. Two cliffs, at the two numbers in an argparse default and nowhere else in 481 columns of ocean; inside, skill decays with distance from each trained edge and bottoms at ≈ 0.17 rather than vanishing. The head is `e032xl_u1_s0.pt` — sunflower-144 at 1024×16, seed 0 — from run #356, staged in the repository as `ml/paper/roll_356.json`; the pixels carry no coordinates in that file, and their geography is recovered from `data/amoc_eval_mask.json`, which the evaluator writes from the same `ys/xs` arrays it scores. This is *one seed*: the figure shows *where* skill sits, not a level.

far to the east of the band. Denominator size does not order skill here.

What that costs the numbers in this section. 25.0% of window pixels, 23.9% of corridor pixels and 30.2% of the RAPID section lie inside the block, so every rolled score in Table 5 is a blend of trained- and untrained-longitude skill. The direction is deflationary: including the block makes every number *worse* than its trained-longitude counterpart, so nothing reported here is inflated by it. The *stencil advantage* is a different matter. At h=6 the 144-point head beats the frozen no-neighbour gate by +0.091 over the corridor; decomposed by the block, that is +0.182 on the 24% of corridor pixels inside it and +0.062 on the 76% outside, so 48% **of the advantage is earned on pixels the model was never trained on** — over the whole window, 59%. The split is stable across the four stencil heads rolled here (42–48% corridor, 57–59% window). That is a defensible thing for a stencil to be good at — reading neighbours is precisely how one covers a hole in one's own coordinate input, and the neighbours of an untrained pixel include trained ones. But *spatial coupling helps forecasting* and *spatial coupling helps a model extrapolate into a hole in its own coordinate input* are different claims, and no number in this report separates them. About half of the measured stencil advantage belongs to the second.

The corridor paradox dissolves. The previous revision of this figure reported that the corridor is *not* the skilful part of the window and could not say why. Split by the block, it is: outside the block the corridor and the rest of the window are indistinguishable (0.864 against 0.866), and the whole of the corridor's apparent deficit is inside it (0.229 against 0.369). The corridor is not selecting unskilful pixels. Fast-current pixels inside an untrained longitude block are simply the hardest thing in the window to extrapolate to — which is what one would expect if the information the model is missing there is advective. Every contrast quoted in this subsection reproduces on the arm's second seed to within 0.02; the one quantity that does not is a level rather than a contrast (the block-centre value reads 0.171 and 0.148 on the two seeds), and nothing here is quoted as a level. One seed is a picture, not a number.

What would settle the diagnosis beyond argument is cheap and not yet done: retrain one stage-2 head with a *different* held-out longitude block and confirm that the band moves with it. It is not a one-line change, which is why it is not in this revision — the stage-2 trainer reads the block from its frozen codec's saved arguments (`ck["args"]["holdout_lon"]`) and has no switch of its own, so either the plumbing grows a stage-2 knob or the codec is retrained too. We record the prediction here rather than the result.

8.2 An inversion at base scale, and the audit that cleared the instrument

At the then-standard stage-2 size (576×8, 34M parameters), a factorial over stencil geometries produced a clean and disturbing result: across eleven arms the one-step and rolled scoreboards were *anti-correlated*. The widest, farthest-reaching stencils posted the best forecast ratios in the programme and rolled *worse than the no-neighbour baseline*; the best rolled arm was a minimal ring of 8 points at 222 km whose one-step forecast was mediocre. The full base-scale table is preserved in Appendix A.

Because an inversion of that shape could equally be a bug in the evaluator, it was audited before being believed ("investigate thoroughly and not hypothesize"). Six artifact classes were tested and eliminated on the archived evaluations alone, zero GPU spent: per-head scored-set differences (sample counts are identical to the pixel), broken rolled-input construction (at h=1 the roll consumes observed truth and reproduces the training ordering exactly), baseline choice (ordering invariant under climatology, persistence, and damped persistence), amplitude collapse (amplitude ratios do not track the inversion — the far-reach heads predict confident, full-amplitude, increasingly *wrong* patterns), horizon weighting (the top ranks are invariant), and boundary/dead-slot artifacts — eliminated with the sharpest measurement of the audit: the rolled penalty *rises* with the fraction of the stencil that is live ($r = +0.176$, monotone in quintiles), so pixels whose far stencils are fully over ocean pay most. An instrumented re-roll added channel structure: the penalty concentrates on the slow 300–700 m heat and salt reservoir — the AMOC-relevant fields — while the same wide stencils *keep* their advantage on fast wind-stress channels under iteration. The instrument was clean; the phenomenon lived in the model-under-iteration.

8.3 Capacity breaks the inversion

The resolving experiment was a 2×2 — head size (576×8 vs 768×12) $\times$ input width (34 vs 55 sunflower points), three seeds per cell, everything else frozen. On the forecast ratio the capacity main effect is -0.029 paired at every seed ($\sim 30\times$ the three-seed noise bar); the width effect is -0.002, same sign at every seed; and there is *no interaction* — width helps by the same small amount at both sizes. Capacity is worth three times the entire geometry axis of Appendix A.

The decisive question was whether that gain *rolls* — a one-step gain that evaporated under iteration would have been exposure bias wearing a scaling law's clothes. It rolls: paired by seed, the 768×12 tier gains **+0.050** corridor AUC over 576×8 on the same 55-point sunflower (+0.052/+0.048), and the same geometry that rolled at 0.569–0.573 (below the no-neighbour 0.589) rolls at 0.621–0.622 —

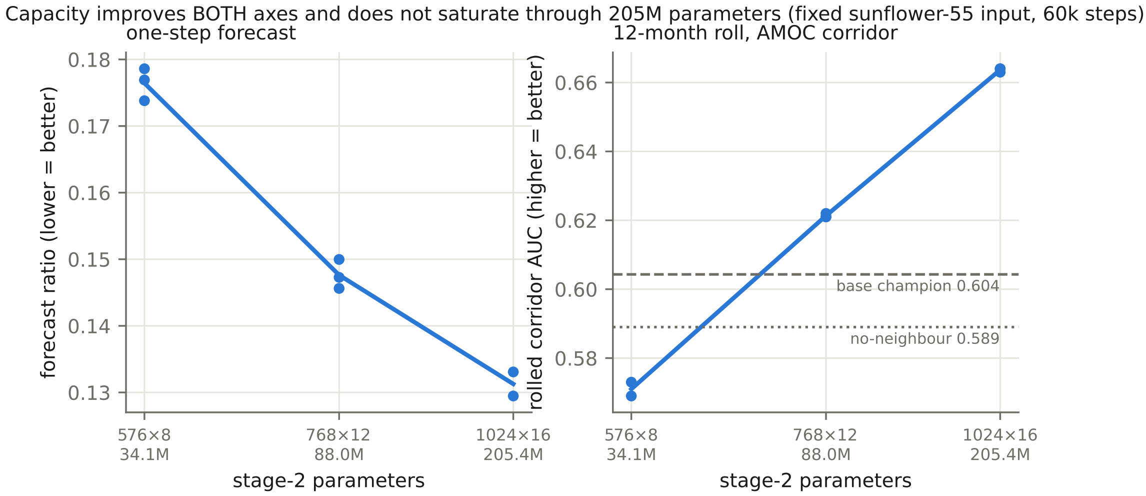

Figure 8: The capacity ladder at fixed input (sunflower-55, 60k steps), all seeds shown, parameters measured from the checkpoints. Left: the one-step forecast ratio. Right: the rolled corridor AUC, against the no-neighbour baseline (dotted) and the best base-scale geometry (dashed). The 576×8 tier is *below* both reference lines — that is the inversion of §8.2 — and the same input geometry crosses both lines the moment the head is large enough. Neither panel shows saturation at 205M.

past the no-neighbour baseline *and* past the base-scale champion's 0.604. The E-026 inversion was a capacity-starvation artifact: a 34M head fed 55×64-dimensional inputs learns a one-step mapping whose errors compound under iteration; an 88M head learns one whose errors do not. The channel fingerprint of the audit (slow subsurface pays, wind survives) is what that starved mapping looks like.

One tier up (1024×16, 205.4M, same input, same steps), both scoreboards improve again: forecast ratio 0.1295–0.1331 (the project's best), corridor AUC **0.664**/0.663/0.664 — +0.042 over the 88M tier, +0.075 over the no-neighbour baseline, with a seed spread of 0.001 against the forecast axis's 0.004: at this capacity the rolled skill is essentially deterministic given the data. Fig. 8 shows the ladder; Table 5 is the per-seed record. No saturation is visible on either axis through 205M parameters.

8.4 The other axes: width and step budget

With capacity established as the first-order variable, the remaining axes were measured against it (Fig. 9). **Width**: at the 88M tier, sunflower inputs of 34, 55, and 89 points give forecast ratios of $0.1498 \to 0.1476 \to 0.1453$ (means of three seeds; paired per rung ≈ -0.002, same sign at every seed) — small, consistent, unsaturated. **Step budget**: extending the 89-point arms from 60k to 120k to 200k steps under a horizon-free exponential-decay schedule traces $0.1449 \to 0.1377 \to 0.1361$ (two-seed means), i.e. a per-step value that falls $\sim 6\times$ between the first extension and the second. The comparison that matters: the 88M/89-point model at its *full* 200k budget (0.1355–0.1366) still does not reach the 205M model at 60k (0.1295–0.1331). At this scale, *parameters beat steps*, which is the same verdict the corridor gave.

Where width really stops. Combining the widest inputs with the largest head settles the width question on the forecast axis and re-opens it on the roll. At 1024×16 and 200k steps, 89 and 144 points give forecast ratios of 0.12362/0.12158 and 0.12358/0.12146 — paired differences of -0.00004 and -0.00012, roughly $17\times$ inside the seed spread. On the one-step scoreboard width is *finished* at 89 points, and we said so.

The corridor disagrees, and it is the scoreboard that matters. Paired by seed, xl144 rolls at 0.68067/0.67558 against xl89's 0.67433/0.67058: +0.0063 and +0.0050, same sign at both seeds, and the ladder $0.6637 \to 0.6725 \to 0.6781$ across $55 \to 89 \to 144$ points is monotone with no visible flattening. The effect is comparable to the tier's own seed spread (0.004–0.005 here, against the 0.001 we measured at xl55), so we read it as consistent and suggestive rather than decisive — but its *direction* is not in doubt, and it is the opposite of what the forecast ratio predicted. **The decision this reverses is**

head	tier	s0	s1	s2	mean
no neighbours (gate)	576 × 8	0.589	—	—	0.5890
base55 (sunflower-55)	576 × 8	0.569	0.573	—	0.5710
ring-8@222 (base champion)	576 × 8	0.608	0.599	0.606	0.6043
big55 (sunflower-55)	768 × 12	0.621	0.621	0.622	0.6213
big34 (sunflower-34)	768 × 12	0.628	0.618	0.625	0.6237
xl55 (sunflower-55)	1024 × 16	0.664	0.663	0.664	0.6637
xl89 (sunflower-89)	1024 × 16	0.674	0.671	—	0.6725
xl144 (sunflower-144)	1024 × 16	**0.681**	**0.676**	—	**0.6781**
xl233 (sunflower-233)	1024 × 16	0.675	0.673	—	0.6739
not a capacity change — same 88M tier, different input or objective					
sun89, 60k steps	768 × 12	—	0.639	0.630	0.6345
sun89, **200k** steps	768 × 12	—	0.639	0.630	0.6341
ring-8@222 (near-field)	768 × 12	0.646	0.661	0.661	0.6560
znoise, σ=0.7 (sun-55)	768 × 12	**0.683**	**0.674**	—	**0.6785**
the composition — noise × input width, at the 205M tier					
znoise × xl144	1024 × 16	**0.722**	**0.725**	—	**0.7235**
znoise × xl233	1024 × 16	**0.725**	**0.723**	—	**0.7240**

Table 5: **Rolled corridor AUC, per seed** — the reference table for the metric defined in §2. Every row is read from an evaluation whose gate head reproduced its pinned 0.643 exactly; the full per-horizon and per-channel outputs are archived per evaluation run (`ml-metrics` branch). The upper block is the capacity ladder at fixed input. **The middle block is the result that reorders this section**: neither row is a capacity change, and both beat the tier they belong to — input noise by +0.057 and near-field geometry by +0.035 over the same 88M model, with the *noised 88M* model beating the *clean 205M* one outright. **The bottom block is what happens when the two are composed**, and it holds the best numbers in the programme by a margin larger than the gap between the top of the capacity ladder and its second rung: 0.7235/0.7240 against 0.6785 for noise alone and 0.6781 for width alone. The two noised rows sit 0.0005 apart across a 1.6× change in input width — which is the finding of §8.5, in one line. The clean xl233 row is now a completed pair: its seed-0 partner, lost at first to a truncated checkpoint upload, was retrained and rolled, and the two seeds read 0.67492/0.67292 — a spread of 0.0020, the tightest of the five pairs at this tier, and a mean 0.0042 below xl144. The base-scale geometry table behind the ring-8@222 row is Appendix A.

a concrete one: a previous revision concluded from the forecast axis alone that a 233-point stencil should not be built. That conclusion is withdrawn. Saturation on a one-step metric is not evidence of saturation under iteration, and this programme has now made that mistake in both directions — E-010 closed the capacity axis in a regime that could not exercise it, and E-032 closed the width axis on a scoreboard that could not see it.

The 233-point rung, and where the ladder actually stops. The stencil that decision built has now been rolled at *both* seeds, and it answers the question against the extrapolation. xl233 rolls at 0.67492/0.67292 — pair mean 0.6739, seed spread 0.0020 — against xl144's 0.68067/0.67558, mean 0.6781. Paired by seed the difference is −0.0058 and −0.0026; on the means it is −0.0042, where the +0.005/rung trend predicted ≈ 0.683. **On the clean arm, width beyond 144 points buys nothing measurable.** That is what the pair licenses and it is deliberately not more. The −0.0042 is about twice the sd this tier's replicate record assigns to a difference of two paired means (0.0021 pooled, 7 dof): enough to exclude +0.005/rung, which is the hypothesis that was on the table, and not enough to say that width *hurts*. The ladder 0.6637 → 0.6725 → 0.6781 → 0.6739 across 55 → 89 → 144 → 233 points stops climbing at 144, and the earlier n=1 reading of this rung — which could distinguish flat from +0.005 but not flat from slightly down — is retired in favour of the pair.

The confound this arm was designed to exclude does not apply. Slot occupancy was measured before dispatch and is scale-invariant across a 7× range of point counts — 47.6%, 47.4%, 47.1%, 47.0% and 46.9% of corridor slots resolve to live ocean at 34, 55, 89, 144 and 233 points — so a null at 233 cannot be read as "the window ran out". More points bought proportionally more live input, as at every rung below, and bought nothing. Width and window have not become the same axis; width has simply stopped paying.

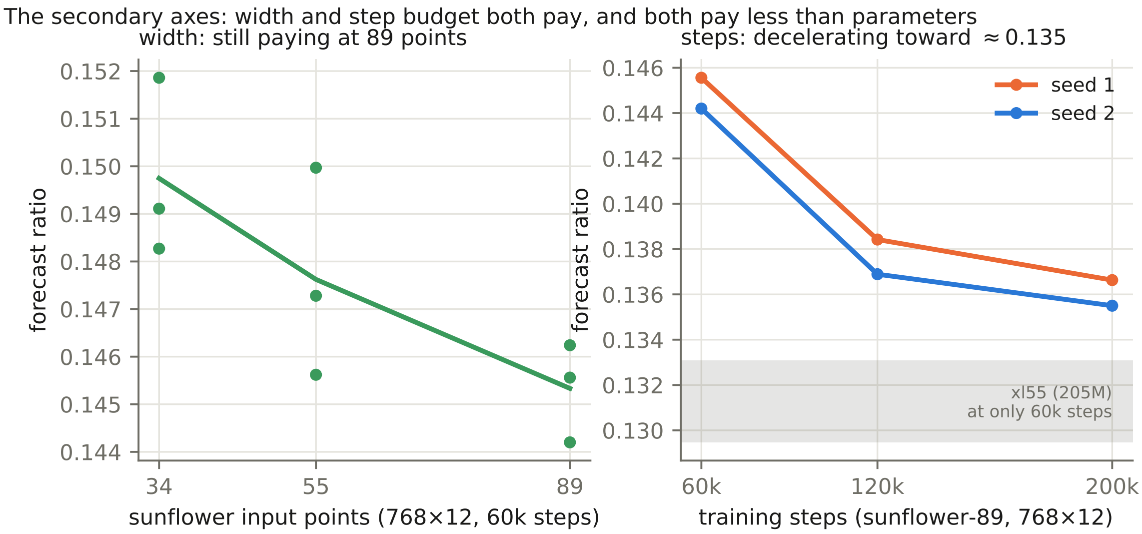

Figure 9: Left: input width at the 88M tier — the forecast ratio improves by ≈ 0.002 per Fibonacci rung and has not saturated at 89 points. Right: step budget at 89 points — the first 60k-step extension buys -0.0072 paired, the second (80k steps) only -0.0016; the trajectory is bending toward $\approx$0.135, and the 205M tier at 60k steps (grey band) remains out of reach of the 88M tier at any step budget measured.

Both axes, rolled at 88M. The same split appears one tier down (gate 0.643, five of five heads). **Width does roll**: at the 88M tier, 89 points give corridor AUC 0.6345 against 0.6213 for 55 and 0.6237 for 34 — +0.013, several times the tier's ±0.005 seed spread, and in the same direction as the forecast axis. **Steps do not.** Paired by seed, the 200k arms roll at 0.63867 and 0.62958 against the 60k arms' 0.63883 and 0.63017: differences of −0.0002 and −0.0006, both *negative*, both an order of magnitude inside seed noise. Three and a third times the training compute bought a decisive −0.0088 on the one-step forecast and *nothing at all* on the twelve-month roll.

The per-horizon breakdown says why, and it is not a flat null. The 200k model is *better* than the 60k model at short range (seed 1, h=1: 0.761 vs 0.756) and *worse* at long range (h=9: 0.602 vs 0.607), with the crossover near h=5; the extra training sharpened the one-step mapping at the cost of its behaviour under iteration, and the corridor's flat average is those two effects cancelling. That is the same trade the noise experiment of §8.6 attacks from the other end — deliberately *worsening* the one-step fit to buy iteration robustness, and gaining +0.057 for it. Extra steps are the anti-noise intervention, and the roll prices them accordingly.

8.5 Noise × width: one axis carries all of it

Every axis in §8.4 was measured alone. Composing the two that appeared to pay — input noise and the widest stencil, at the largest capacity — produces the programme's best rolled numbers, and a sharper result than the one we expected: across the full 2×2, *noise carries the entire gain and width contributes nothing.*

input	clean	noised (σ=0.7)	Δ noise
sunflower-144	0.6781	**0.7235**	+0.0454
sunflower-233	0.6739	**0.7240**	+0.0501
Δ width	−0.0042	+0.0005	

Table 6: **The 2×2, rolled corridor AUC.** Both rows at 1024×16, 200k steps, same schedule; every cell is now the mean of two seeds. The noise column is +0.045 to +0.050; the width row is −0.004 and +0.0005. The interaction term — whether noise pays *more* at greater width — is +0.0047, at the scale of the seed spread it would have to clear.

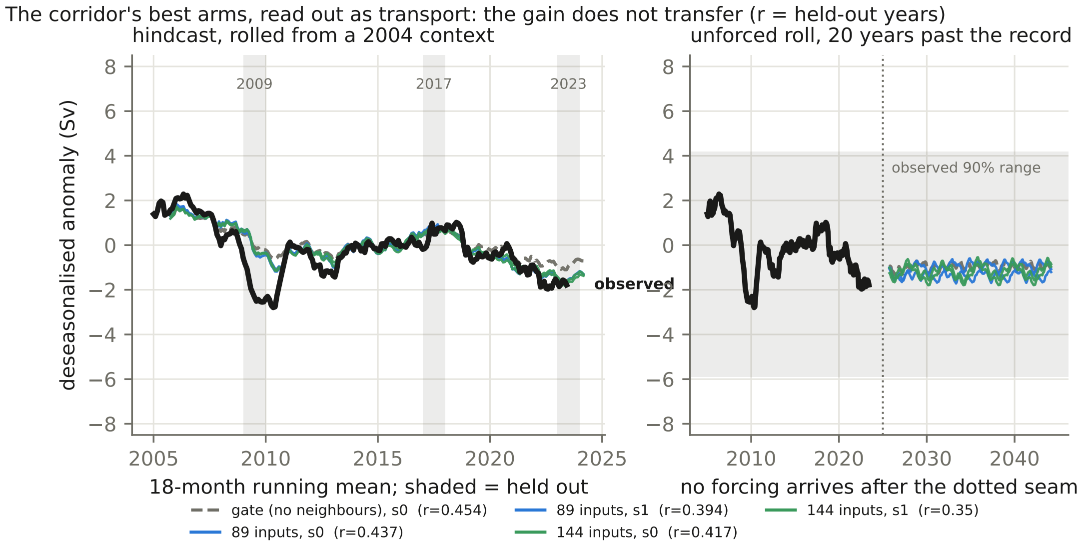

Figure 10: The AMOC read-out of the two widest 205M arms, rolled. Left: hindcast from a 2004 context, with the three never-trained years shaded — those columns are the honest test, because outside them the roll replays months the codec saw in pretraining. Right: twenty years of unforced roll past the end of the record. The flatness on the right is the finding, not a forecast: with no wind or heat-flux forcing after the seam, the learned dynamics relaxes toward climatology and stops moving, which is a damped attractor rather than a quiet ocean. Interactive version, with the per-arm table, at `blauewelt.github.io/earth/ml/paper/figs/amoc_roll.html`.

Table 6 is the whole experiment, and every cell of it is a seed pair. Noise buys +0.0454 at 144 points and +0.0501 at 233; width buys −0.0042 clean and +0.0005 noised. The interaction term is +0.0047, which is the size of the seed spread it would need to exceed, so we do not claim it. Read across the noised column instead: 0.7235 and 0.7240, five ten-thousandths apart, across a 1.6× change in the number of input points. **Once noise is on, the stencil width stops mattering entirely.**

A correction, recorded. On the 233-point pair alone this section claimed a *compounding* interaction — that noise paid where width had just stopped paying. The 144-point pair, evaluated a few hours later, shows that reading was an artifact of having one cell of a 2×2: noise pays the same amount at both widths, so there is nothing for width to compound with. The claim was wrong in the direction this programme is most prone to, and §10 keeps the general form of the lesson: a single arm of a factorial design licenses a main effect and never an interaction.

The comparison is as clean as this programme gets: *same* seed, *same* 1024×16 head, *same* 233-point sunflower input, *same* 200k steps under the same schedule, differing in one flag. Both arms are now two-seed, so the 233-point comparison pairs seed-for-seed twice over: σ=0.7 noise rolls at 0.72517/0.72292 against the clean arm's 0.67492/0.67292 — **+0.0503** at seed 0 and **+0.0500** at seed 1, the same number to three decimals, and ten times the largest seed-pair delta measured anywhere at this tier (0.0051). Each pair is itself tight (0.0023 noised, 0.0020 clean). The 144-point pair reads 0.7220/0.7250 against a clean 0.6781: +0.045, the same effect at a different width.

Two reference points make the size legible. Noise alone at 88M reached 0.6785; width alone at 205M reached 0.6781; the composition reaches **0.7235/0.7240**. The whole capacity ladder, from the no-neighbour gate to the widest clean model, spans 0.589 → 0.678; one flag, applied at the top of it, adds more than half that span again.

The effect is also strikingly *constant*. Input noise bought +0.057 at 88M on a 55-point stencil, +0.045 at 205M on 144 points and +0.050 at 205M on 233 points — a spread of 0.012 across a 2.3× change in capacity and a 4.2× change in input width. Whatever exposure bias costs this architecture, it costs about the same everywhere we have looked, and correcting it returns about the same everywhere. That is a more useful shape than a compounding one: it makes noise a fixed dividend available at any point on the ladder, rather than a reward for having climbed it.

horizon h	clean (mean)	noised (mean)	Δ
1	0.775	0.769	−0.006
2	0.730	0.747	+0.017
4	0.692	0.730	+0.038
6	0.665	0.722	+0.057
8	0.651	0.717	+0.066
10	0.635	0.707	+0.072
12	0.627	0.701	+0.074

Table 7: **Where the noise gain lives.** Corridor MSSS against climatology by horizon, for the 233-point 205M arms with and without σ=0.7 input noise. Both columns are two-seed means. The noised model is *worse* at one step and better at every horizon after, with the advantage growing at every horizon shown and reaching +0.074 at a year (its maximum over all twelve is +0.077, at eleven months). This is the exposure-bias signature stated as a prediction in §8.6, measured: noise buys stability under iteration and pays for it in one-step accuracy.

Table 7 is the mechanism. The noised model is *behind* at h=1 by 0.006 and ahead at every horizon thereafter, and the gap widens across the table — +0.017 at two months, +0.038 at four, +0.074 at a year. Nothing about that shape is a better one-step mapping; it is a model that degrades more slowly when fed its own output, which is exactly and only what the intervention was designed to produce. The prediction registered in §8.6 was that noise would cost the forecast ratio and buy the roll. It costs 0.006 on the one-step corridor and buys 0.074 at twelve months.

It also settles a sequencing question that was left open deliberately. A σ sweep was judged not worth running until it was known whether noise still paid at 205M, on the grounds that $\sigma = 0.7$ was set from the model's own measured one-step error and never tuned. It pays, at the top of the ladder, at a size no capacity increase has matched — so the sweep is now the obvious next experiment, and the untuned value it starts from is already the best configuration this programme has.

The honest caveat. σ was set from the *clean* model's one-step error. The noised model's error is larger, which means the value that was self-consistent at 88M is not self-consistent here, and the direction of the mismatch is unknown. The composition is measured; its optimality is not claimed.

What the corridor gain does *not* buy. Fig. 10 rolls these same two arms out as transport, and the result is a caution against reading corridor AUC as AMOC skill. Over the three held-out years the no-neighbour gate reads $r = +0.45$, xl89 reads +0.44/+0.39 and xl144 reads +0.42/+0.35 — if anything a mild *decline* with the input width that gains most on the corridor, and a seed spread (0.05–0.07) that swamps every difference between arms. The corridor ranking does not transfer to the transport read-out at all.

§8.5 sharpens this to the point of discomfort. The noised 233-point arms gain +0.050 corridor AUC over their own clean control — the largest single effect in the programme — and their banded transport correlations are 0.460/0.352/0.451 and 0.457/0.341/0.422 against the clean arm's 0.460/0.350/0.443: equal at short range, equal or slightly worse at long. The *no-neighbour 32M gate*, which rolls the corridor at 0.589 and sees no neighbourhood at all, still reads the best transport bands of any head evaluated (0.470/0.375/0.492). A +0.135 spread in rolled field skill produces no ordering whatsoever in transport. Whatever the corridor is measuring, twelve months of it is not the overturning. Two readings are available and we cannot yet separate them: the corridor scores the model's own input fields where the transport probe scores a thin integral of them, so the gain may be real and simply not where the probe looks; or the wide arms are winning the corridor on fields the transport does not depend on. Either way, §10(6) stands unchanged, and the bridge from rolled field skill to transport skill at range remains the programme's open problem rather than something scale has quietly solved.

8.6 Exposure bias: two mechanisms under test

The audit's finding — errors that compound under iteration — suggests the classic exposure-bias remedies, and two are under test. *Input-noise training* adds Gaussian noise to the input embeddings at the model's own measured one-step error scale ($\sigma = 0.7$ z-units), so training-time context statistically resembles roll-time context; as predicted it *costs* one-step forecast (0.1548/ 0.1539 vs. the clean tier's 0.1476). Whether it buys the roll was open when this section was first written and is now answered: §8.5 measures +0.050 corridor AUC at the 205M tier, paired at both seeds, with the gain confined to $h \geq 2$ and reaching +0.074 at twelve months. This is the one exposure-bias mechanism the programme has confirmed. *One-hop wide unroll* exploits a dependency-cone observation: to train through one step of self-generated context at stencil width S, one does not need to roll the whole ocean — only the S input pixels of the next step, each of which is exactly a depth-1 prediction from its own observed window. This is implemented (detached depth-1 forwards over the slot pixels, an assembled $t{+}1$ window, a differentiable second step against Z_{t+2} at half weight) and verified by alignment tests; its first two arms were stopped mid-run by an operator decision after the measured step cost came in at 8.5× the clean model (gather-dominated), and the checkpoint state is preserved for a decision on resumption. Neither mechanism has yet produced a rolled verdict; both are falsifiable against the clean 88M tier's corridor AUC of 0.6213.

8.7 The dependency cone, and what every rolled number assumes

A stencil head predicts a pixel from its S neighbours, so a two-step prediction depends on the neighbours' neighbours, and the set of pixels an h-step prediction depends on — its *cone* — grows by the stencil's reach every step. At 4444 km that is $4444h$ km against Earth's 40,075 km circumference, so the cone wraps the planet within a few steps while our window is a basin. The evaluator already rolls every one of the 84,405 window ocean pixels, which is the most it can do; the question is whether the window is large enough.

Measured rather than argued (pure geometry over the real neighbour table, seeded from the RAPID section): for the 144-point sunflower the cone covers 26.9% of the window at h=1, 99.5% at h=2 and **100% at h=3**, after which **∼50% of every step's requested neighbours resolve to land or off-window**. The near-field ring-8@222 reaches only 30.7% of the window by h=12 with 4.8% of its demand unmet — so it is *reach*, not slot count, that escapes.

This is not an error: dead slots are encoded as zero in training exactly as in evaluation, so no number here is invalidated. But it means every rolled result in this report carries a boundary condition we had not stated — **the world outside the window is held at its climatological mean** — and that the fraction of a model's input owned by that assumption grows with its reach. §8.1 adds the second boundary condition of the same kind, on the other side of the input: a quarter of the pixels these numbers are averaged over lie in a longitude block no stage of the model was trained on, and roughly half the stencil's rolled advantage is earned there. Both caveats reward reach for the same reason, and neither was stated in the artefacts of the evaluations reported here. It is the most plausible single explanation for §8.8's two surprises: a wide-and-far model rolls on a state that is half boundary assumption, a near-field one barely depends on the boundary, and input noise makes a model robust to exactly the kind of corrupted context the assumption produces. It also promotes a global tensor from a data-volume argument to a correctness requirement for long rolls at wide reach.

8.8 Standings

Table 8 is the current state of the programme, and the rolled column has just overturned the reading this section was drafted with. Capacity dominates the *forecast* axis and pays handsomely on the roll — both of those stand. What is no longer true is that capacity dominates *everything*: at a fixed 88M, adding Gaussian noise to the model's own inputs is worth +0.057 corridor AUC and beats the 205M model, while a near-field 8-point stencil is worth +0.035 and also beats every wide-input arm of its own tier. The two cheapest interventions in the programme outrank its most expensive one.

The honest summary is therefore a rebalanced one. On one-step forecast, scale is the whole story and the axes are ordered parameters ≫ width > steps. On the twelve-month roll, *robustness to the model's own error* is at least as valuable as parameters; the ranking of input geometries does not follow the forecast ranking at all — the two scoreboards disagree again (§8.2) at a capacity where starvation cannot explain it — and the step-budget axis, worth −0.0088 on the forecast, is worth *zero* rolled (§8.4). Of the four axes measured at fixed 88M capacity, only two survive iteration: width, weakly (+0.013), and error robustness, decisively (+0.057). §8.7 gives what we think is the underlying reason. One qualification belongs on every corridor-AUC comparison in this table and is quantified in §8.1: about half of a stencil head's rolled advantage over the no-neighbour gate is earned on the quarter of the

arm	tier	params	input pts	forecast ratio	corridor AUC
xl144 (200k)	1024 × 16	208.9M	144	0.12358/0.12146	**0.6781**
xl89 (200k)	1024 × 16	206.9M	89	**0.12362/0.12158**	0.6725
xl55 (60k)	1024 × 16	205.4M	55	0.1295–0.1331	0.6637
sun89 (200k)	768 × 12	89.7M	89	0.1355–0.1366	0.6341
sun89 (120k)	768 × 12	89.7M	89	0.1369–0.1384	—
sun89 (60k)	768 × 12	89.7M	89	0.1442–0.1462	0.6345
big55 (60k)	768 × 12	88.0M	55	0.1456–0.1500	0.6213
big34 (60k)	768 × 12	86.9M	34	0.1483–0.1519	0.6237
big-ring222 (60k)	768 × 12	—	8	0.1509–0.1557	**0.6560**
znoise-55 (60k)	768 × 12	88.0M	55	0.1539–0.1548	**0.6785**
base55 (60k)	576 × 8	34.1M	55	0.1738–0.1786	0.5710
ring-8@222 (base)	576 × 8	—	8	0.1848	0.6043
no neighbours (gate)	576 × 8	—	0	—	0.5890

Table 8: **Stage-2 standings, both scoreboards**, 16 August 2026. Forecast ratio ranges are seed min–max; corridor AUC is the seed mean, every value behind a passed gate. "In flight" marks evaluations running at the time of writing. The two scoreboards now *agree* at adequate capacity — the anti-correlation of Appendix A is confined to the 34M tier.

window whose longitudes no stage of the model was ever trained on. The advantage is real and the ranking is unaffected — every arm here is scored on the same pixels — but part of what the stencil is being rewarded for is covering a hole in the model's own coordinate input rather than forecasting the ocean. The next experiments are the compositions nobody has run: noise × 205M, noise × near-field, and a sweep of σ, which was set from the model's measured one-step error and never tuned.

9 The static-channel lesson

An ablation found the two climatology input channels contribute *exactly nothing* to any metric: a static map is a function of position, the encoder receives position, and so the channel carries no information the model does not already have. The general rule — a channel is worth what position and season cannot explain of it — now gates every data-acquisition decision. Climatological knowledge is still essential, but it enters as preprocessing arithmetic (the anomaly transform), never as a feature the network must carry through its bottleneck.

10 Limitations and open questions

(1) The wind-only baseline bounds honest claims: monthly AMOC skill is substantially Ekman, though the unpooled head now clears the baseline by +0.10–0.16 where the pooled probes had shrunk the margin to noise. (2) The head result is new and strong, and strength is when scrutiny matters most. The second codec seed has now been run and it removed the pretraining margin (§7.3); what remains outstanding is a paired significance test, and the fact that the head has been run only on RAPID — the MOVE and low-frequency versions may move too. The head is also demonstrably the noisiest probe we own (seed spread 0.036 against the ridge's 0.012), so single-seed head numbers should not be quoted anywhere, including by us. (3) d_z=64 was chosen on a 12-channel tensor and carried unchanged into 24–25 channels; the sweep has not been repeated at the new width. (4) Absolute k-fold values are mildly optimistic — the codec saw non-holdout months' *fields* during self-supervised pretraining (transports were never seen by anything); codec-to-codec comparisons are unaffected. (5) The rollout's horizon is measured only to the holdout-year boundary; the crossover to no skill lies beyond 12 months and is not yet located. (6) Transport forecasting at range is unsolved. (7) *The held-out longitude block is scored anyway.* 45°W–25°W is excluded from training in both stages and included in every rolled evaluation, unrecorded in the artefacts of every evaluation in this report (§8.1, Fig. 7); the evaluator now emits the split, but the numbers here predate it. Its effect on the reported levels is deflationary, so no published number is inflated; its effect on the stencil *contrast* is not, and about half of that contrast is earned inside the block. The decisive confirmation — retrain with a different block and watch the band move — needs a stage-2 holdout knob the trainer does not have and has not been run. (8) *A*

methodological casualty worth recording: the stage-2 results for two runs were withdrawn after we found their embedding cache had been populated by a different codec — the shape check (T, P, d_z) matched, so training proceeded happily on the wrong representation, and the signature was healthy embedding-space skill beside catastrophic decoded skill. The cache key now carries a hash of the codec's weights, so a stale cache is a miss rather than a lie. We report the affected numbers as withdrawn rather than regenerating them, because both codecs belong to the superseded coarse tensor.

For the rolled-forecast programme of §8, four things are open at the time of writing, all mechanical rather than conceptual: the rolled corridor evaluations of the width, step-budget, and exposure-bias arms are in flight (their forecast-axis numbers are final); the widest-input × largest-head runs (89 and 144 points at 205M, 200k steps) land within hours; the one-hop unroll arms await a resume decision; and the capacity ladder has no measured top — every tier so far improved both scoreboards, and the next tier is the programme's standing next experiment. The corridor AUC itself carries one structural caveat: it scores the model's own input fields rolled forward, not transport — the bridge from rolled field skill to transport skill at range is the same open problem as (6) below.

The training-time ceiling has now been located directly: a 1,000,000-step run at pilot size (LR decayed to a constant 10% floor by 50k steps, probes every 50k) completed in 2.6 h on one RTX 4090 and is *flat from the first probe to the last* — linear r oscillates in $[0.43, 0.48]$, channel-space skill vs. persistence is pinned at 32%, and the final checkpoint's k-fold (0.571) sits within seed noise of the 40k-step run. Twenty-five times the compute of the pilot regime buys nothing at 0.92M parameters: the pilot is capacity-limited, which is what the Chinchilla arithmetic predicted (∼13M parameters for this tensor) and what the 10M-parameter codecs now training are for. Next levers, in expected-value order: paired significance for the head results; the head probe at MOVE and at low frequency; re-running the d_z sweep at C=24; then the approved data ladder — (a) extending the time axis to 1982, where monthly SST and wind exist pre-GLORYS and the Florida cable's 1982–92 decade becomes truth, (b) 0.25° resolution for the GLORYS-derived channels (16× the pixels), (c) doubling the Argo pressure levels — with the Chinchilla arithmetic redone and the codec re-sized at each rung; and label-efficiency curves over the 792 non-RAPID truth months.

A Base-scale results superseded by capacity

Everything in this appendix was measured at the 576 × 8 (34M) stage-2 tier, and §8.3 showed that tier to be capacity-starved for wide inputs: the geometry ranking below is a property of a too-small head, not of the ocean. It is preserved because the numbers are real, the discipline that produced them (pre-registered hypotheses, paired seeds, gate reproduction) is the same as the main text's, and because a ranking this clean that *dissolved* under one scaling step is itself a finding worth keeping in view.

A.1 The geometry factorial at 34M

Eleven stencil arrangements, three seeds each where completed, rolled corridor AUC (every evaluation gate-passed):

stencil	pts	reach km	AUC mean	vs champion
ring-8 @222	8	222	**0.6043**	0
spiral-8	8	890	0.5883	−0.016
no neighbours	0	0	0.5890	−0.015
three rings narrow	12	1000	0.589	−0.015
spiral-13	13	1000	0.5863	−0.018
two rings 8+8	16	555	0.5857	−0.019
three rings wide	24	1000	0.5813	−0.023
spiral-24	24	4444	0.578	−0.026
elliptic-24	24	4444	0.568	−0.036
spiral-34	34	4444	0.566	−0.038
ring-16 @222 (n=1)	16	222	0.564	−0.040
sunflower-34	34	4444	0.553	−0.051

The ranking is monotone in slot count within the spiral family (8 → 13 → 24 → 34 gives 0.588 → 0.586 → 0.578 → 0.566), and two isolations pin the axes: doubling points at fixed 222 km radius costs

−0.044; widening reach at fixed 8 points costs −0.016. Meanwhile the *forecast* ordering of the same arms ran the other way — the sunflower and deep-spiral arms were the best one-step models in the programme at the time. That anti-correlation, audited clean in §8.2, is what §8.3 resolved: at 88M and 205M the wide sunflower geometry wins *both* scoreboards, and the production-geometry question was re-opened at scale rather than settled by this table. The base-scale champion's ring-8@222 configuration is itself being re-evaluated at the 88M tier at the time of writing.

A.2 Early stage-2 capacity sweep and codec training curves

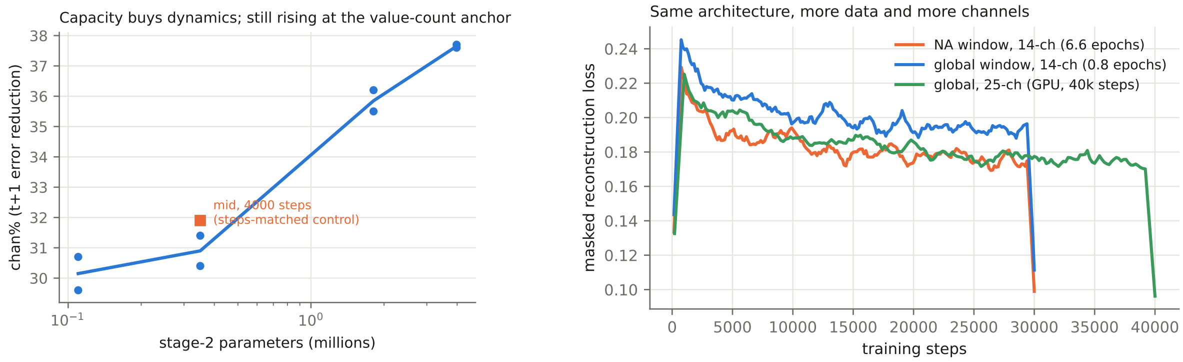

Figure 11: Left: the original stage-2 capacity sweep (0.11–4M parameters, 12-channel coarse tensor, no stencil) — the first sign, in retrospect, that stage-2 capacity was the binding axis; it was still rising at its top end. Right: codec training curves across generations.

The original stage-2 capacity sweep (Fig. 11, left) was run on the coarse tensor at 0.11–4M parameters and read, at the time, as "capacity buys dynamics." It deserved more weight than it got: an early experiment at one slot and small K ("E-010") found stage-2 skill insensitive to capacity and closed the axis; that negative was real *in its regime* (a 1-slot input barely exercises a large head) and wrong outside it, and re-opening the axis at K=24 × 55 slots is what produced §8.3. A settled negative is settled only in the regime that settled it.

References

[1] E. Frajka-Williams (2015). Estimating the Atlantic overturning at 26°N using satellite altimetry and cable measurements. *GRL* 42.

[2] A. Sanchez-Franks et al. (2021). A dynamically based method for estimating the AMOC at 26°N from satellite altimetry. *Ocean Science* 17.

[3] A. Solodoch et al. (2023). Machine learning-derived inference of the meridional overturning circulation from satellite-observable variables. *JAMES* 15.

[4] Y. Meng et al. (2024). Circumpolar transport and overturning strength inferred from satellite observables using deep learning. *JAMES* 16.

[5] E. Wölker et al. (2025). Estimating the AMOC from Argo profiles with machine learning trained on ocean simulations. *Ocean Science* 21.

[6] L. Goddard et al. (2013). A verification framework for interannual-to-decadal predictions experiments. *Climate Dynamics* 40.

[7] F. Lehmann et al. (2026). Frozen foundation-model backbones with shallow decoders as an evaluation standard. *JGR-MLC.*

[8] E. Worthington et al. (2021). A 30-year reconstruction of the Atlantic meridional overturning circulation shows no decline. *Ocean Science* 17.

[9] L. Caesar et al. (2018). Observed fingerprint of a weakening Atlantic Ocean overturning circulation. *Nature* 556.

[10] Y.-G. Ham, J.-H. Kim, J.-J. Luo (2019). Deep learning for multi-year ENSO forecasts. *Nature* 573.

[11] J. Taylor, M. Feng (2022). A deep learning model for forecasting global monthly mean sea surface temperature anomalies. *Frontiers in Climate* 4.

[12] K. Bi et al. (2023). Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619.

[13] R. Lam et al. (2023). Learning skillful medium-range global weather forecasting. *Science* 382.

[14] K. He et al. (2022). Masked autoencoders are scalable vision learners. *CVPR*.

[15] T. Nguyen et al. (2023). ClimaX: a foundation model for weather and climate. *ICML*.

[16] C. Bodnar et al. (2024). Aurora: a foundation model of the atmosphere. arXiv:2405.13063.

[17] J. Jakubik et al. (2023). Foundation models for generalist geospatial artificial intelligence (Prithvi). arXiv:2310.18660.

[18] Y. Cong et al. (2022). SatMAE: pre-training transformers for temporal and multi-spectral satellite imagery. *NeurIPS*.

[19] C. Brown, M. Kazmierski, V. Pasquarella et al. (2025). AlphaEarth Foundations: an embedding field model for accurate and efficient global mapping from sparse label data. arXiv:2507.22291.

[20] J. Hoffmann et al. (2022). Training compute-optimal large language models. arXiv:2203.15556.

[21] D. Roemmich, J. Gilson (2009). The 2004–2008 mean and annual cycle of temperature, salinity, and steric height in the global ocean from the Argo Program. *Progress in Oceanography* 82.

[22] E. Kalnay et al. (1996). The NCEP/NCAR 40-year reanalysis project. *BAMS* 77.

[23] B. Huang et al. (2021). Improvements of the daily optimum interpolation sea surface temperature (DOISST) version 2.1. *Journal of Climate* 34.

[24] R. Adler et al. (2018). The Global Precipitation Climatology Project (GPCP) monthly analysis: new release 2.3. *Atmosphere* 9.

[25] G. McCarthy et al. (2015). Measuring the Atlantic meridional overturning circulation at 26°N. *Progress in Oceanography* 130.

[26] C. Meinen, M. Baringer, R. Garcia (2010). Florida Current transport variability: an analysis of annual and longer-period signals. *Deep-Sea Research I* 57.

[27] U. Send, M. Lankhorst, T. Kanzow (2011). Observation of decadal change in the Atlantic meridional overturning circulation using 10 years of continuous transport data. *GRL* 38.

[28] M. S. Lozier et al. (2019). A sea change in our view of overturning in the subpolar North Atlantic. *Science* 363.

[29] M. Kersalé et al. (2020). Highly variable upper and abyssal overturning cells in the South Atlantic. *Science Advances* 6.

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